Relaxing the steady-state assumption displaces forest soil carbon models along the trade-off between litter input and turnover

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Abstract

Soil carbon models used in national greenhouse gas inventories are commonly started from a soil in balance with its litter input, although the soils of managed forests are rarely in balance. We calibrated six soil carbon models, the three versions of Yasso, the model used in the Finnish inventory, and three simpler benchmark models with one to three pools, against soil carbon measured in three campaigns between 1986 and 2024 on 456 permanent plots of the Finnish National Forest Inventory. Each model started from a soil in balance in 1917 and was run forward along a reconstructed history of litter input. The litter input and the input of 1917 were estimated together with the model parameters in a Bayesian calibration, with an error model in which observations share part of their error by plot, by region and by campaign. All six models represented the national trajectory of soil carbon, and none could be distinguished from the others by its fit. The calibration placed the models at a litter input 1.3 to 1.7 times the input supplied by a tree-litter product and, in Yasso07 and Yasso15, at mean transit times 27 and 11 per cent shorter than published, whereas Yasso20 was not displaced. Litter input and transit time trade off against each other, and returning to the published transit times cost little in fit, so the observations suggest a direction rather than impose a value. The position on this trade-off changed the projection of Yasso07 only. Under a constant litter input the models placed the soil 26 to 42 per cent below its state of balance, consistent with an earlier reconstruction, while the projected sink fell by a third to a half within 60 years; the size of that sink depended on how depleted the soil was estimated to be at the start. None of the models represented the differences between individual plots, and about half of what the models left unexplained could be recovered from site variables that already exist. The study provides a baseline for further development of the models used in forest soil carbon accounting. [Check the length against the journal's limit, and revisit once the Limitations and the discussion of the starting state are settled.]

1 Introduction

1.1 A soil that is still catching up with its forest

Forest soils are the largest carbon (C) store of the forest ecosystem. In the boreal zone specifically, more of the C held by a forest lies in its soil than in its trees (Pan et al., 2011).

But what is most crucial, more than the snapshot of total C stocks, are the trajectories: if soils are gaining or losing C. And usually a national greenhouse gas inventory focus on these, and reports the annual change in the stock rather than the stock itself. A soil that is still gaining C and a soil that has stopped gaining C (which is defined as equilibrium, since there is still fluxes in and fluxes

out that just balance each other) can hold the same amount of C, and the two soils differ only in where they have come from and where they are going. The trajectory is therefore what defines if a reported sink will persist, and for how long.

Such trajectory is influenced by the history of the soils considered, even land use changes relatively far in time. While litter falls to the soil continuously and most of its C returns to the atmosphere within a few years, a small but consistent fraction of that C joins material that decomposes on a scale of decades to centuries. Because this slower C persists for so long, a forest whose management changed even a century ago can stand on a soil that is still adjusting to such change. Such a soil, not in equilibrium, is gaining or losing C at a certain rate.

1.2 The soil term of a national inventory is often modelled rather than measured

Countries determine the soil organic C (SOC) in their inventories in more than one way. Some measure it. Sweden estimates the change in the C stock of its forest mineral soils from a soil inventory carried on the permanent plots of its national forest inventory since 1993, on a ten-year cycle, and reports the measured difference (Swedish Environmental Protection Agency, 2026). Germany does the same from two national forest soil inventories two decades apart (Grüneberg et al., 2014). Repeating a national soil survey often enough to yield a stock change is expensive, however, and the more common route is a decomposition model driven by the forest inventory: Finland, Norway, Switzerland and Austria run Yasso07 on their inventory plots (Statistics Finland, 2025; Norwegian Environment Agency, 2025; Didion et al., 2016), and Canada runs CBM-CFS3 (Kurz et al., 2009). But the modelling route implies a difficulty that Norway's inventory states plainly, that "in a managed forest landscape neither forest biomass, soil nor DOM pools can be assumed to be in an equilibrium state" (Norwegian Environment Agency, 2025). And yet any model needs to start from a certain state, and usually that state is determined by assuming equilibrium conditions at some point past in time. This means assuming a certain mean input level as constant, and calculating the C stocks from that assumption, before starting a transient simulation.

Finland, specifically, reports to the United Nations Framework Convention on Climate Change through a chain in which forest inventory data and biomass models supply an estimate of the litter falling to the soil, and the decomposition model Yasso07 converts that litter estimate into an annual change in the soil C stock, for dead organic matter and soil organic matter combined to a depth of one metre (Statistics Finland, 2024), following the method of Liski et al. (2006). No measurement of soil C enters the Finnish chain at any point.

1.3 What the Finnish estimates have reported, and why

The quantity produced in that way has been, for most of the reporting period, a sink, and a large one. The combined pool of dead organic matter and soil organic matter on mineral forest land was reported as a sink of about 10 million tonnes of carbon dioxide equivalent per year at the beginning of the 2000s (Natural Resources Institute Finland, 2025). In the 2024 submission the same pool was still a sink of 4.8 million tonnes in 2022, against a living biomass sink of 12.6 million tonnes (Statistics Finland, 2024), although the soil term carries a reported uncertainty of 31.5% against 16.3% for the biomass term (Lehtonen and Heikkinen, 2016). In the 2025 submission, which recalculated the litter input from the thirteenth National Forest Inventory, the mineral soils of forest land turned "from a net sink to a net source from 2021 onwards", emitting 0.4 million tonnes of carbon dioxide in 2023

(Statistics Finland, 2025). The same submission reports that the C sink of Finnish forest land has fallen by more than 80 per cent since 1990, and that the land-use (LULUCF) sector as a whole has become a net source (Statistics Finland, 2025). A term of the size and uncertainty of SOC can decide the sign of such a balance.

The 2025 submission names the decrease of the litter input as one of the reasons for the turn of the soil term to a source, and traces that decrease mainly to a lower estimate of foliage biomass (Statistics Finland, 2025); the Luke release on the same figures adds global warming (Natural Resources Institute Finland, 2025). The model used for those estimates is started from a soil in balance with the litter input of the sixth National Forest Inventory (1971–1976); such input rose for decades with the growth of the standing biomass and with the residues of an increasing volume of fellings, and the modelled soil accumulated C in response. The estimated litter input then fell, mainly through a lower estimate of foliage biomass, and the modelled soil began to lose C (Statistics Finland, 2025). The reported soil term is therefore, in both directions, the response of a modelled soil to a litter input that departs from the level the soil was started in balance with. Whether the real soil was in balance with its litter input in the 1970s, and how far it has come since, is not something this modeling framework can tell, also because there are no soil measurement to inform the model and everything follows from model assumptions. [one clause or one panel: the reported term per hectare against the measured rate, on the CRT series — decide after the colleagues confirm the −10 figure]

1.4 Three campaigns on the same plots

But the permanent plots of the Finnish National Forest Inventory have been sampled now three times: a first campaign whose fieldwork ran from 1986 to 1995, the European BioSoil programme in 2006, and a third campaign in 2024. On the plots measured in all three campaigns, the mean soil organic carbon (SOC) stock rose from 64.5 tonnes of C per hectare to 71.5 and then to 73.6, a gain of 7.0 tonnes over the first interval and a gain of 2.1 tonnes over the second (each plot counts equally in the above means, so the means describe the average sampled plot). The inventory samples the north of the country at a third of the density used in the south, and each northern plot therefore stands for three times as much forest land. Weighted by the area of forest that each plot represents, the mean stock rose from 63.7 to 68.8 and then to 69.9 tonnes of C per hectare. The gain is smaller when the plots are weighted by area, because the southern plots gained more carbon than the northern ones, but in both cases the gain over the second interval was much smaller than the gain over the first. Three points give already a trajectory, that can be used to constrain a SOC model.

The measured soil has been gaining C, as the inventory reports, but it seems that the gain has been recently slowing down.

1.5 Steady state as a starting condition

Any SOC model needs to be initialized somewhere. The model has to be told how much C the soil already holds on the day the simulation begins, and how that C is divided between fractions which decompose at different speeds, for a soil whose history was never observed.

The standard solution is steady-state initialisation: the initial state is taken to be the state at which the C stock would stabilize (zero change) under a litter input and a climate held fixed. That state can be obtained by solving the model equations directly or by running the model forward until the stock settles. The choice between the two is a matter of implementation and what matters is

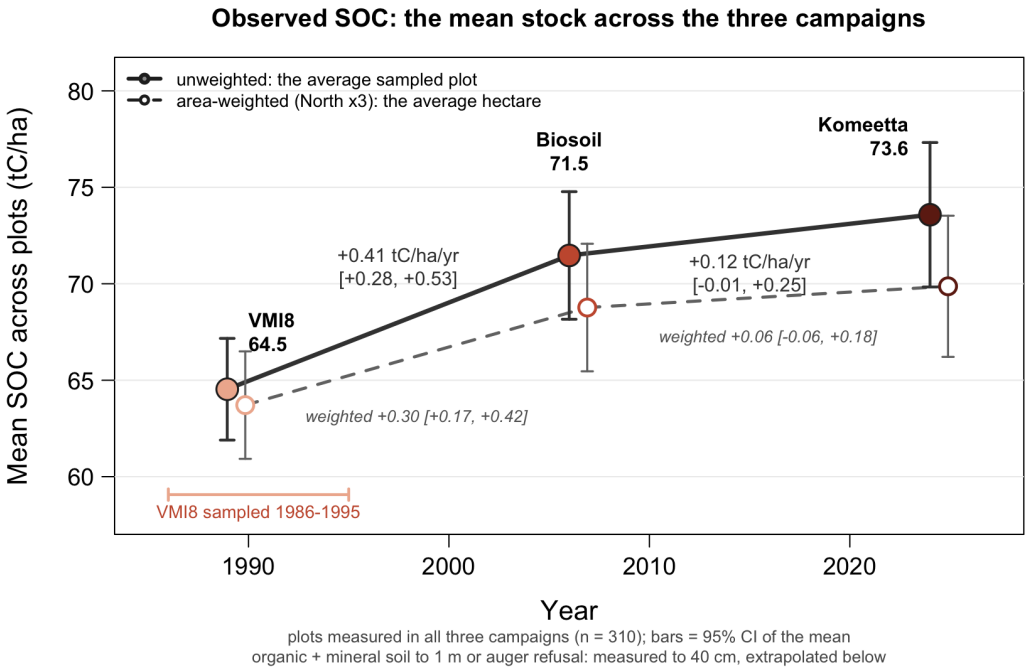


Figure 1: The measured soil carbon stock of Finnish forest plots in three campaigns. Each point is the mean stock across the plots measured in all three campaigns ($n = 310$), and the bars show the 95% confidence interval of that mean. The first campaign is drawn at the mean year in which its plots were actually sampled, and the bar low in the panel shows that the sampling of the first campaign ran from 1986 to 1995. Each segment is labelled with the rate of change over that interval and with the 95% confidence interval of the rate, calculated from the paired difference between campaigns on the same plots. The filled points and the solid line are the unweighted means, in which every plot counts equally; the unweighted means are the ones compared with the models, whose trajectories are also averaged over plots with equal weight. The open points and the dashed line are the means weighted by the area of forest that each plot represents, with a weight of three for the plots in the north, which are sampled at a third of the southern density; the weighted means describe the average hectare of forest on mineral soil. The stock is the whole soil profile: the organic layer and the mineral soil, measured to 40 cm and extrapolated below, down to one metre or to the depth at which the auger stopped. The stock rose over both intervals, and the rise over the second interval was smaller than the rise over the first.

the core assumption: that the SOC was in balance with its litter input on the day the simulation began.

Most model families used for national reporting of forest soil C are initialized in that way. The nationwide assessment that evaluated Yasso07 and ROMULv against Finnish soil measurements initialised both models at steady state (Lehtonen et al., 2016). The spin-up procedures of global land models rest on the same assumption (Luo et al., 2016), as do applications of the model RothC (Coleman and Jenkinson, 1996; Contreras et al., 2026). The Canadian forest C model CBM-CFS3 looks at first like an exception, because the model builds its starting state from repeated cycles of growth and disturbance (Kurz et al., 2009); but a cycle around a constant average is still representing a trajectory that can be approximated as flat, and cannot represent any other trajectory.

In the Finnish inventory, the initial state is obtained by running the model under an unchanging litter input until the C stock stops changing, using the average litter input of the sixth National Forest Inventory (1971–1976) and the average weather of 1960–1990, and the justification offered is that "approximately ten years of simulation since spin-up is enough to cancel out the effect of the spin-up level" (§6.4, citing Statistics Finland, 2024; Peltoniemi et al., 2006).

There is already considerable uncertainty around that assumption. Peltoniemi et al. (2006) repeated the Finnish calculation as a Monte Carlo experiment and found the uncertainty of the soil C sink to be "dominated by the soil model initialization, but the effect decreased with time": the initial state contributed 77 per cent of the standard deviation of the annual sink in the first year, 30 per cent after ten years, and 21 per cent after sixteen. But the initial state whose uncertainty was propagated was itself a steady state, so what the experiment measured is the spread around an assumed level rather than the full uncertainty in the trajectory.

Wutzler and Reichstein (2007), working with the Yasso model, examined the common practice of choosing model parameters so that the modelled steady state matches an observed C stock. A soil that is in fact still accumulating implies an equilibrium stock far above the stock observed, so fitting under the assumption of balance systematically overestimates the decay rate of the slowest C pool, and overestimates the stock of a recently disturbed site. The same conclusion was drawn, in the same year, in a review of seven soil C models for national accounting written largely by the group that built the Finnish chain (Peltoniemi et al., 2007): the assumption of a soil in balance with its current inputs "is likely to be violated in most applications", and a model whose parameters are set so that its equilibrium matches a measured stock needs "a further correction of the parameters and the pools" if the accumulation over a century is not to be underestimated. The steady state assumption biases the parameters retrieved by model–data fusion and not only the initial state (Carvalhais et al., 2008), and later work has measured what the choice of initial state costs in predictive error (Lee and Viscarra Rossel, 2020; Kanari et al., 2022).

The closest predecessor of the present study is Liski et al. (2006), and the two studies share the conceptual approach of pushing the steady state assumption as far back in time as possible. Liski and coauthors simulated the soil C of Finland's forests forward from 1922 to 2004, over nearly the same period, with the same model family, and from litter inputs derived from the same National Forest Inventory. But Liski and coauthors had no repeated measurement of the same soil, from which trajectory we could constrain an initial state, so they assumed one, calculating the pools at 1922 "by assuming a steady state with mean litter input between 1922 and 1936 and mean temperature between 1901 and 1930". The present study keeps their forward simulation, but replaces that one assumption with an estimate that draws information from the measured SOC trajectory.

1.6 A managed forest landscape is not an equilibrium landscape

A forest whose standing biomass has changed within the time the soil needs to adjust to a new litter input leaves a soil that is not in balance with that input, and this is a rather common occurrence. Finland is a well-documented case of one. The Finnish forest of the early twentieth century had been heavily used: slash-and-burn cultivation had passed over most of the productive forest land of eastern Finland, grazing in the forest suppressed regeneration, and tar burning and the selective removal of the best timber had drawn down the accessible stands. By the first national forest inventory old trees were effectively absent from southern Finland outside the least accessible areas, and the volume of wood in Finnish forests had reached its historical low point (Aakala et al., 2023). The recovery that followed was not steady. Logging fell during the war years and the growing stock rose in the 1940s, but the use of wood for reconstruction and export reduced it again through the 1950s and 1960s, and the strong increase in logging before the fourth inventory is visible in that inventory as a slowdown of the increment (Korhonen et al., 2021). The pressure had three sources: the war reparations paid to the Soviet Union until 1952, about four per cent of the national product each year, delivered as industrial goods but financed by an economy whose exports came from the forest sector (Mitrunen, 2024); the loss in 1944 of a ceded territory that held more than a tenth of the country's forests and a fifth of its pulp capacity (Kochetkova, 2020); and the resettlement of the evacuated population on land taken from the forest. Only after the late 1960s did the growing stock rise without interruption, from roughly 1.4 to 2.6 billion cubic metres over the century as a whole, an increase of 84 per cent (Korhonen et al., 2024) (Figure 2).

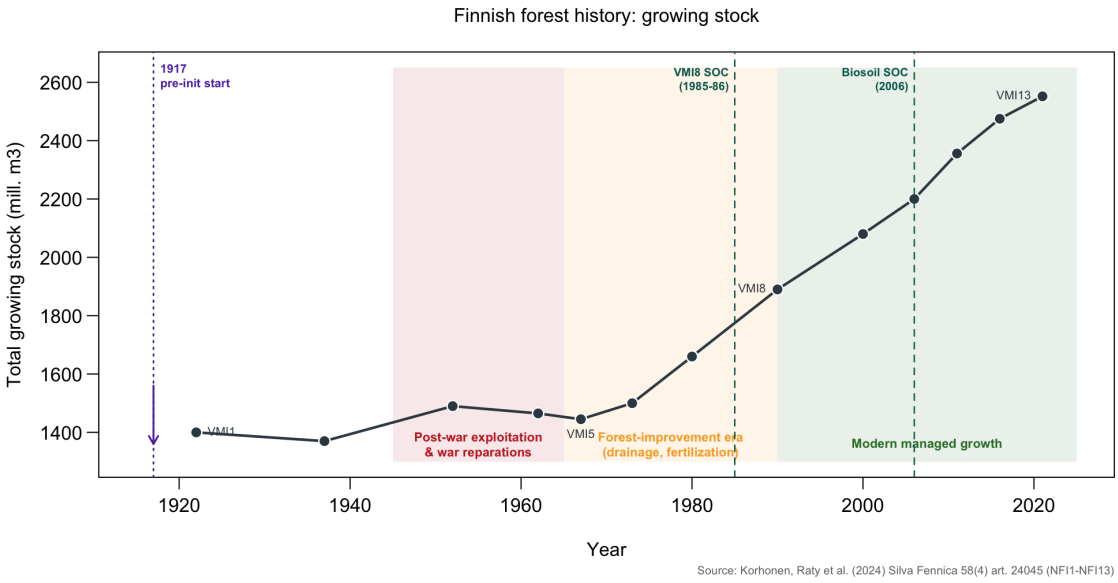


Figure 2: The growing stock of Finnish forests, 1921–2023. Growing stock rose by 84 per cent over the century shown, and most of the rise occurred after the late 1960s. The litter falling to the soil is tied to the standing biomass, so the litter input that the soil has integrated has been rising throughout the period that the models are asked to reproduce. Data from Korhonen et al. (2024).

But soil C responds slowly to a change in litter input. The pools that turn over within years adjust within a few decades, whereas the slow pool that holds most of the stock approaches its new level only over centuries, so that a soil disturbed several centuries earlier can still be accumulating

(Wutzler and Reichstein, 2007). The move from a C-poor soil under a low litter input to a richer soil under a higher one is therefore measured in decades to centuries, not years. That time is much longer than the mean time a unit of carbon spends in the soil, because the carbon leaving the soil comes mostly from the fast pools while the carbon stored in it sits mostly in the slow one; the two timescales are distinct properties of the same model (Sierra et al., 2018).

The consequence for the starting state of a soil model was stated in this literature two decades ago.

Peltoniemi et al. (2004), working from soil C stocks measured onlyonce on Finnish stands of different ages, showed that such a chronosequence cannot separate the two processes that act on the soil at the same time: the change in soil C over the course of a rotation, driven by management, and the slower recovery of the whole landscape from its historical depletion.

Liski et al. (2006) approached the long-term trend by simulation, and stated both its direction and its shape: Finnish forest soils would go on accumulating C even if the litter input were held at its 2004 level, and, centuries later, the stocks would stabilise at a level about 38 per cent above that of 1922. An accumulation that stabilises is an accumulation that slows dowhn, and the slowing is the part of that statement which has received the least attention since.

Finland now has three soil C campaigns on the same plots, spanning roughly four decades, which measures the trend directly, and that trajectgory is the information that can make the initial state something to be estimated rather than assumed.

1.7 What a soil started in balance cannot do

Two quantities govern the C stock of a decomposition model: the litter input the soil receives, and the mean transit time of C in the soil. A measured stock does not separate the two, because the same stock is reached by a large litter input decomposed quickly and by a small litter input decomposed slowly. A soil in balance holds a stock equal to the product of the two quantities, $C^* = J/k$ for a single pool (Olson, 1963), so the product is all that a stock can fix. The same observation is reproduced by many combinations of input and kinetics, which is the situation Beven (2006) named equifinality, and holds for every soil C model and for the transient phase of a simulation as much as for its steady state. A trajectory narrows that set, but only through its shape. For a single pool the ratio of the rate of change in one interval to the rate in the interval before depends on the decomposition rate alone, and not on the litter input nor on how far from balance the soil started, so the variation of such rate over time carries additional information.

A soil started in balance is not unable to accumulate or to slow down, it still responds to the litter input and the climate after the start. What it cannot carry is a disequilibrium that already existed on the start date. With decomposition rates held at their published values, there are limits to the shape of the trajectories the model can represent and the model would not be able to fully represent a trend that originates also in an initial disequilibriu,. But if the rates are freed so that a balanced start fits the observed trajectory anyway, the missing history would be absorbed into the rates (Wutzler and Reichstein, 2007), and this might result in biased future extrapolations.

1.8 Objectives

This study revises two assumptions about the starting state, and the two work together.

First, the state is estimated rather than assumed: the litter input at which the soil is taken to be in balance is a free parameter, and the measured trajectory is what informs it, because a stock alone cannot.

Second, the date of that balance is pushed back to 1917, as far as the forest record allows, so that whatever is wrong with the assumption of balance in 1917 (and something will likely be, since a real steady state is something achieved only in unmanaged, primary forests) has sixty-eight years of simulated history to dissipate that error while being influenced by the transient simulation before the first measurement.

A starting state assumed in balance just a decade before the first measurement has neither protection. This makes possible to answer the questions below.

I) Once the starting state is estimated from the measured trajectory, where does that trajectory place the models on the ridge, in parameter space (and defined by similarly good fit to the data), between litter input and transit time, relative to their published parameterisations? And how far does it narrow that exchange?

II) Does the position on such ridge matter beyond the observed window? Do two calibrations that reproduce the same measurements from different positions extrapolate differently, and for which model structures?

III) What part of the variation between individual plots (due to local conditions) remains unexplained, and what information would the model need to explain it?

2 Materials and methods

2.1 Study plots and soil carbon observations

The study is based on data from permanent forest plots of the Finnish National Forest Inventory for which soil carbon has been measured in three campaigns: the eighth national forest inventory (VMI8), the BioSoil campaign of 2006, and the Komeetta campaign of 2024. After the exclusions described below, 456 plots are available for calibration, carrying 1205 plot-year observations of soil carbon in total. Of the 456 plots, 365 plots (80 per cent, carrying 959 observations) were used to fit the models, and 91 plots (20 per cent, carrying 246 observations) were held out and used only for evaluation. Plots were assigned to the two groups at random, and a plot was assigned as a whole, so that no plot contributes observations to both groups.

The first campaign is not a single year. Its plots were sampled region by region between 1986 and 1995: 69 plots in 1986, 76 in 1987, 103 in 1988, 54 in 1989 and 72 in 1995. Averaged over the plots, the sampling year of the first campaign is 1989, and that is the year at which the campaign is drawn whenever it is shown as a single point. In the calibration, each observation is compared with the model in the year in which that plot was actually sampled.

On the plots measured in all three campaigns (310 plots, the set on which every observed rate of change in this paper is computed), the mean stock of the groups sampled in 1986, 1987, 1988, 1989 and 1995 differs by 18.3 tonnes of carbon per hectare at first visit, which might reflect a trend within that same campaign. The five groups of plots defined by their first-campaign sampling year (1986, 1987, 1988, 1989 and 1995) differ by 37.2 tonnes in 2006 and by 34.2 tonnes in 2024, years in which every plot of every group was measured in the same campaign. The difference between the five groups is therefore a property of which plots each group contains, and not of the year in which

it was sampled, because the difference is larger when the sampling year is held constant.

2.2 Soil carbon stocks on a single homogenised basis

Since the three campaigns were measured and processed under protocols that differed in ways that matter for a comparison between them, all three campaigns were rebuilt on a uniform basis, starting from the layer-by-layer carbon stocks measured in each of the three campaigns and applying one set of conventions to all three. Those stocks are corrected for the volume occupied by stones and for the bulk density of the fine earth. The volume of stones was measured in 2006 on every plot by the rod-penetration method of Viro (1952), in which a steel rod is driven into the mineral soil at the sampling points and the mean depth it reaches is converted into a stone volume, and the same stone volume is applied to the first campaign (Lehtonen et al., 2016; Kramarenko, 2012). The bulk density of the mineral layers was predicted from the equations of Tamminen and Starr (1994) for all layers of the first campaign and for most layers of 2006, and measured directly for part of the 2006 samples from the upper 20 cm (Kramarenko, 2012). [Confirm that the 2024 campaign uses the same stone volumes and the same bulk density equations.] The coarse-fragment correction matters because the coarse-fragment content of these plots has a median of 43 per cent by volume, so a mineral stock computed without it is too large by a factor of about 1.6.

The quantity used as the calibration target is the whole soil profile: the measured organic layer, plus the measured mineral soil to 40 cm, plus a modelled deep tail, integrated only as far down as the auger actually reached on that plot and in no case deeper than one metre. The depth to which the profile is integrated is therefore a per-plot quantity and not a fixed one metre, since using a fixed depth of, for example, one metre would invent soil carbon below bedrock on the 38 plots where augering stopped at 10, 20 or 40 cm. The modelled deep tail was validated against the measured 40–80 cm layer of the BioSoil campaign, which was held out of the fit: the median ratio of predicted to observed carbon in that layer is 0.96, with a bias of −2.5 tonnes of carbon per hectare.

The rebuilt target was checked against the national figures published for the two campaigns for which such figures exist. Applying the same plot exclusions as the official calculation, and integrating over the same layers, the rebuilt target returns weighted national means of 59.06 and 61.05 tonnes of carbon per hectare for 2006 and 2024 against official values of 59.06 and 61.05, that is, agreement to the nearest 0.01 tonnes per hectare on 446 plots.

Two further steps were taken with the first campaign, both of them established from the structure of the measurements themselves rather than from the fit. First, the layers of the three campaigns were standardised, because they were recorded differently: the organic layer of the first campaign contains the fermentation and humus horizons only, whereas the organic layer of the two later campaigns also contains the litter and moss layer. Each plot's own litter and moss layer from 2006 was therefore added to its first-campaign organic layer, so that the same compartment is compared across the three campaigns. Second, the first campaign is treated as what it is, a campaign whose fieldwork ran over several years rather than a campaign carried out in a single year. The year in which each plot was actually sampled is on record, and those years, rather than one nominal year for the whole campaign, are the years at which the models are compared with the observations.

[Data availability and acknowledgement: the soil carbon measurements are not a published dataset, so the statement has to say where they come from, who compiled them and what of the compiled form can be deposited. Settle this with the people who hold them before submission.]

On this final basis the median whole-profile stock is 62.0 tonnes of carbon per hectare for the

first campaign, 66.4 for 2006 and 67.2 for 2024, and the observations vary between plots with a coefficient of variation of 0.44 to 0.53 depending on the campaign.

2.3 Litter input

Litter input is taken from the annual reconstruction of tree biomass and litter input for the permanent plots of the Finnish National Forest Inventory by Ťupek et al. (2026), published as an open dataset. In that reconstruction, the tree biomass measured by the inventory is interpolated to every year with the help of satellite-derived basal area, and litter is computed as the biomass of each tree component multiplied by a turnover rate for that component, plus the branches, foliage, stumps and roots left on site when a drop in basal area identifies a harvest; the full method is given in Ťupek et al. (2026). The product supplies, for each plot and year, the carbon in litter from trees, separated into non-woody litter (foliage and fine roots), fine woody litter (branches and coarse roots) and coarse woody litter (stumps), and each of the three classes is further divided into the four chemical fractions used by the Yasso models, using proportions specific to each tree species and component. The product does not contain a separate term for trees that die without being harvested, and it does not include the understorey vegetation, because the product is a tree litter product by design. Across the study plots the mean litter input carried by the product is $\bar{J} = 2.511$ tonnes of carbon per hectare per year.

The national greenhouse gas inventory estimates the same flux for the mineral forest soils of southern and northern Finland, from the forest inventory and biomass models (Statistics Finland, 2025). On the part that the two estimates have in common, the litter from the turnover of living trees and from harvest residues, they agree in level in southern Finland, where most of the study plots lie, and neither rises after about 2006; in northern Finland the inventory estimate lies higher and keeps rising [Supplementary figure: S18_litter_vs_inventory]. The inventory estimate also carries the litter of trees that die without being harvested, a term that grows over the period and that the plot product does not contain. [confirm with LUKE that the two regions of the inventory data match the region code of the study plots]

The first year of the product is treated separately. The median litter input of 1985 is 0.060 tonnes of carbon per hectare per year against 1.398 in 1986, a step that the series does not repeat in any later year. Because 1985 is both the first year of the simulation and the year in which the first soil observations fall, the value for 1985 was reconstructed for each plot and each chemical fraction by fitting a linear trend over 1986–1990 and extrapolating one year backwards. The reconstruction places 1985 just below 1986, which is the direction the growing-stock record requires.

2.4 Climate

Climate forcing comes from the gridded daily weather observations of the Finnish Meteorological Institute, interpolated from the station network onto a 10 by 10 km grid by kriging with external drift (Aalto et al., 2016) and extended by the Institute to the present. The grids were obtained through the internal weather database of the Natural Resources Institute Finland, which mirrors them from the Finnish Meteorological Institute. From the grid cell nearest to each plot we took the daily mean, minimum and maximum temperature, precipitation and global radiation for 1961–2025, and each model takes from those series the variables it requires. Because the response of decomposition to climate is not a straight line, the average of the response over the years is not the

response evaluated at the average climate. Averaging the climate first and evaluating the response afterwards biases the climate modifier by 5 to 15 per cent in boreal conditions. The climate modifier is therefore computed for each year and averaged afterwards, in all six models.

2.5 The six models

Six models are used, and their roles are different by class. Yasso is the model this study is focused on, because Yasso is the model the Finnish greenhouse gas inventory runs. All three published versions of Yasso are included: Yasso07, Yasso15 and Yasso20. Alongside the three versions of Yasso we included three simple models with one, two and three carbon pools, denoted SP1, TP2 and TP3. The three simple models are there as a benchmark, since their behaviour is transparent enough so that any result obtained with Yasso can be checked against a structure simple enough to reason about quickly.

Due to their different structures and designs, the number of free parameters differs greatly across the six models, from six free parameters in SP1 to twenty-six in Yasso15 and Yasso20. The free parameters of the three versions of Yasso are the twelve fractions describing transfers between pools, the climate parameters, the parameters describing the effect of the size of woody litter, and the two auxiliary parameters described in Section 2.6.

The decomposition rates of Yasso are not fitted, and are held at their published values. Holding the rates fixed is a deliberate choice to keep the model anchored to its original definitions. Decomposition rates and transfer fractions do not act independently on the carbon stock but are always interacting: a slower rate and a larger transfer into a slow pool produce much the same trajectory, so fitting both together improves usually little and makes the result harder to read. Of the two, the rates are the more fundamental to the definition of each model, because the rate of a pool is what distinguishes that pool from the others. Whether the pools of a soil carbon model can also be identified with measurable fractions of organic matter is a long-standing question in soil carbon modelling, and the answer has shaped how Yasso was built. The first version of Yasso assumed that litter consists of groups of organic compounds, each group decomposing at its own rate regardless of the litter it came from, and defined its decomposition compartments accordingly as extractives, celluloses and lignin-like compounds, followed by two compartments of humus (Liski et al., 2005). Yasso07 kept the same principle and defined its four litter pools by fractions that can be measured on litter by sequential extraction: the fraction soluble in acid, the fraction soluble in water, the fraction soluble in ethanol, and the insoluble remainder (Tuomi et al., 2009). Defining the pools by the chemistry of the litter is what makes Yasso operational, because the chemistry of litter can be measured and supplied to the model directly. The same aim, of matching the pools of a model with fractions that can be isolated in the laboratory, has been pursued for other models, for example through fractionation schemes designed to correspond to the pools of RothC (Zimmermann et al., 2007), and whether fractions isolated in the laboratory correspond to the functional pools of models has been examined at length (von Lützow et al., 2007).

Once carbon has entered the soil, the parameterisation of Yasso moves carbon between the four chemical fractions in every direction. In the published parameterisation of Yasso07, for example, 97 per cent of the carbon decomposed from the insoluble fraction is passed to the acid-soluble fraction, and 99 per cent of the carbon decomposed from the acid-soluble fraction is passed to the water-soluble fraction (Tuomi et al., 2009), which represents the transformation of organic matter by decomposers rather than the decay of a fixed set of compounds. The pools therefore enter the model

defined by the chemistry of the litter, and within the soil they act, as in any other compartmental soil carbon model, as compartments defined by their rates of turnover.

The view of pools as compartments of turnover and the view of soil organic matter as a continuum of quality are less opposed than they appear. A chemical fractionation is itself a discretization: it divides a continuum into a small number of classes, the discretization is made in the laboratory rather than in the model, and where the boundaries between classes fall is set by the method chosen, so that the fractions are defined by the procedure that isolates them (von Lützow et al., 2007). The material being divided is not a set of discrete compounds. Soil organic matter is the product of many processes of decomposition, microbial transformation and recombination, much of that organic matter has no defined molecular structure, and the chemical diversity of soil organic matter is large enough that the diversity can be treated as a continuum (Lehmann and Kleber, 2015).

The theory of continuous quality starts from that continuum: Bosatta and Ågren (1991) describe soil organic matter by a distribution of quality, in which the quality of a unit of carbon determines how fast that unit is decomposed, and decomposition in turn moves the carbon along the continuum (Ågren and Bosatta, 1998). The same idea was formalised independently in the geochemistry of sediments, where models with a few discrete fractions of different reactivity are treated as discretizations of a continuous distribution of reactivity (Boudreau and Ruddick, 1991). On this view, a pool of a compartmental model and a fraction of a laboratory procedure are two discretizations of the same continuum, one made in the model and one made in the laboratory, and in the model the rate of decomposition is what places each pool on the continuum. Changing a rate therefore redefines the pool, whereas the transfer fractions describe how carbon moves between pools that have already been defined. We therefore keep the rates as the published parameterisation set them and calibrate the transfer fractions. In all three versions of Yasso the decomposition rates, together with the fraction of decomposed carbon that enters the humus pool, are fixed at their published values: they are constants of the model and not parameters of the calibration, and no prior is placed on them.

2.6 The uncertainty of the litter input

The litter input is among the least certain of the quantities that drive a soil carbon model. Litter is not measured on the plots but derived from tree biomass through biomass models and turnover rates, parts of the input such as the litter of the understorey are missing from tree-based products by construction, and the input carries year-to-year variation that periodic inventories cannot capture. Uncertainty analyses of the Finnish soil carbon estimate have identified the litter input, together with the initial state of the soil, as the dominant sources of uncertainty (Peltoniemi et al., 2006; Lehtonen and Heikkinen, 2016). Those analyses propagated the uncertainty forward, by sampling the input from assumed error distributions and running the soil model with each sample. We instead estimate the error of the input from the soil observations themselves, together with the parameters of the models, so that the correction applied to the input and the uncertainty of that correction are both learned from the data and carried into every result.

The correction is a single multiplier, σ_{input}, applied to the litter input of the product over the whole simulation. The prior of σ_{input} is log-normal with a centre of 1.08, the ratio of the total litter input to Finnish upland soils estimated by Lehtonen and Heikkinen (2016), which includes the understorey, to the tree litter input of the product used here, 2.70 against 2.511 tonnes of carbon per

hectare per year, and a width of 0.25 on the logarithmic scale. Because the multiplier is one number for the whole country, it corrects the national level of the input and not the input of individual plots, whose errors are carried by the error model described below. To test how far the result depends on the centre of this prior, the calibration was repeated with the centre at 1.27, the ratio implied by the reconstruction of Liski et al. (2006).

The same reasoning is extended to the starting state. The litter input of 1917, which sets the state of the soil at the start of the simulation, is not known either, and is estimated through a second parameter, σ_{init}, the ratio of the litter input of 1917 to the litter input of 1985. The prior of σ_{init} is log-normal with a centre of 0.90, the ratio of the growing stock of Finnish forests in 1917 to that in 1985 (Korhonen et al., 2024) converted to a ratio of litter input with the relation between litter input and growing stock observed in the product, and the same width of 0.25; the reconstruction of Liski et al. (2006), based on the same growing-stock record, implies a similar ratio, about 0.86. Both parameters are estimated together with the model parameters, so that the uncertainty of the input and of the starting state propagates into the initial state, into the simulated trajectory and into the predictive intervals of every model.

2.7 Transient initialisation

Rather than starting each plot from a soil in balance with its present litter input, each draw of the parameter vector starts the soil in 1917 and runs it forward to the first soil observation before the comparison with data begins. The year 1917 is a convention, chosen as the year of Finnish independence and therefore the point from which the national forest record is usually counted (any year starting far enough before the observations for the soil to carry a history would work). The state of the soil in 1917 is obtained conventionally, with a steady state assumption with a constant litter input equal to the litter input of 1917, and the litter input is then raised along a prescribed shape over the sixty-eight years from 1917 to 1985, after which the simulation is driven by the annual litter and climate of the record.

The shape along which the litter input is raised is taken from the reconstruction of the litter input to Finland's forest soils published by Liski et al. (2006), who estimated that input annually from 1922 onwards by combining national forest inventory data with biomass and turnover models. Three decisions were made about that reconstruction. First, only the tree-derived terms are used, meaning tree litter together with harvest residues and the litter of trees that died naturally, because those are the terms closest to the litter product used after 1985; that product represents natural mortality only when a death appears as a loss of basal area, but natural mortality is between 2 and 4 per cent of the tree-derived input in the reconstruction, and leaving it out changes the shape used here by less than 0.04 in any year. The ground vegetation is left out of the shape for the same reason that it is left out after 1985, assuming that it is absorbed by the input multiplier described in Section 2.6, and including it before 1985 but not after would make that multiplier mean two different things on either side of the join. Second, the national totals are converted to a per-hectare basis by dividing by the area of upland forest soil. Liski et al. (2006) report the carbon of upland soils both as a national total and per unit area, and the ratio of the two gives an area of 13.90 million hectares in 1922 and 15.22 million hectares in 2004. We placed the growth between the two values in 1965–1980, the period in which Korhonen et al. (2024) locate most of the expansion of productive forest land. Third, only the shape is taken and not the level: the series is rescaled to run from zero in 1917 to one in 1985, so that the reconstruction supplies only the path between the two endpoints.

The two endpoints are set by the two parameters described in Section 2.6: the input at the end of the path is the litter input of the product, averaged over 1985–1989 for each plot and scaled by σ_{input}, and the input at the start of the path is a fraction σ_{init} of the input at the end.

The construction of the starting state is shown in Figure 3.

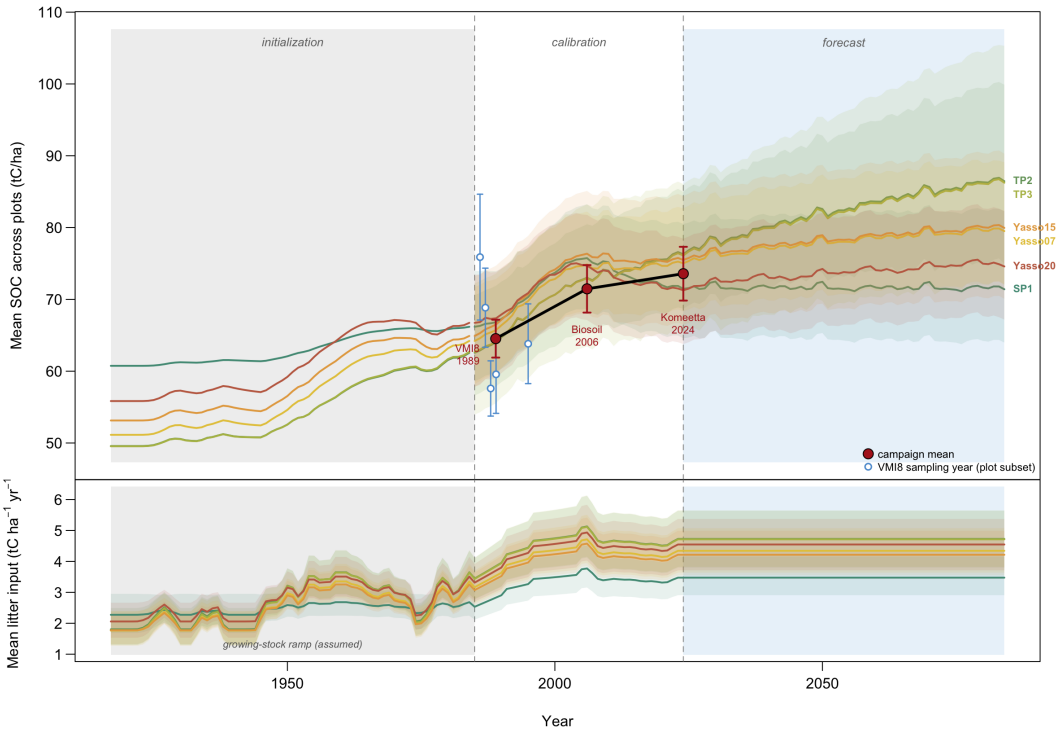


Figure 3: How the starting state of the soil is obtained. Rather than starting each plot from a soil already in balance with its present litter input, the simulation starts in 1917 from a soil in balance with the litter input of 1917, and the litter input is then raised over the following sixty-eight years along a shape taken from the reconstruction of Finland's forest litter input by Liski et al. (2006), after which the simulation is driven by the annual litter and climate of the record. The litter input of 1917 is not known and is estimated as part of the calibration.

2.8 Keeping the models as comparable as possible

The six models cannot be given the same external information. The models differ in structure and in the number of their parameters, and the published literature provides different information for each of them: the three versions of Yasso come with their own calibrations, whereas the three simple models have no calibration for forest soils. The more the models differ in the information they are given, in their assumptions and in their freedom, the less a difference between their results can be attributed to their structure. We therefore reduced that difference as far as the models allow, through one compromise for each class of parameter, summarised in the table at the end of this section. The compromises reduce the asymmetry between the models without removing it, and what remains is stated with each compromise.

2.8.1 Decomposition rates

The rates of the three versions of Yasso are fixed at their published values, for the reasons given in Section 2.5. The three simple models have no rates of their own for forest soils, and take theirs from the ICBM model (Andrén and Kätterer, 1997): the fast rate is fixed, and the slow rate or rates are given a very informative prior centred on the ICBM value, so that the data can move them only slightly. The slow rates are pinned rather than fixed because the ICBM values cannot be transferred to a forest soil unchanged. ICBM was calibrated on an arable long-term experiment, and cropland and forest soils differ in what controls decomposition: cultivation breaks up the soil aggregates that protect organic matter physically in undisturbed soils (Six et al., 2002). Moreover, a rate is only meaningful together with the climate response it was estimated with, and the simple models combine the rates of ICBM with the climate response of Yasso07; the rates were rescaled once so that the two climate responses agree at the climate of the experiment on which ICBM was calibrated, but the two responses differ elsewhere. Fitting the rates freely, on the other hand, would give the simple models a freedom that the versions of Yasso do not have, and a narrow prior around the ICBM value is the compromise between the two.

2.8.2 Transfer fractions

The fractions that describe how carbon passes between pools are free in every model that has them: twelve in each version of Yasso, and the fraction of decomposed carbon passed on to the slower pool in the simple models, of which SP1 has none, TP2 one and TP3 two. All fractions carry a prior of the same width, 0.4 on the logit scale, centred on the published value of each model, which for the simple models is the humification coefficient of ICBM.

2.8.3 Climate response and size of woody litter

The three versions of Yasso take the centres and widths of their climate priors from their own published calibrations (Tuomi et al., 2009, 2011), and [Yasso15/20 source: Yasso20 = Viskari et al. 2022; Yasso15's calibration is unpublished (Järvenpää et al., in preparation) – decide whether to cite the FMI parameter files or the manuscript]. The simple models, which have no published climate response of their own, borrow the climate response of Yasso07, centre and width. Only the versions of Yasso describe how decomposition depends on the size of woody litter, with priors taken from their own calibrations; the simple models have no such term.

2.8.4 The two auxiliary parameters

Every model carries the two parameters described in Section 2.6, σ_{input} and σ_{init}, with the same prior, the same centre and the same width, in every model.

[Parameter-class by prior-criteria table (tab:priors): for each class, the centre source, the width source, the transform and constraint, and which models carry it. It already exists in the methods documentation.]

2.9 Numerical integration

Each model is run forward one year at a time, from 1917 to the last observation, with the litter input and the climate of each year held constant within that year. Within a year, each pool receives its share of the litter input and of the carbon passed on from other pools, and loses a fixed fraction

of its own carbon per unit of time. The carbon in the pools at the end of the year can therefore be obtained in two ways. The simple way is to compute how fast each pool is changing at the start of the year and to assume that the pool keeps changing at that speed for the whole year. The exact way is to use the solution of the equations over the year, which for a single pool is known in closed form: the pool approaches the value at which gains and losses balance along an exponential curve, and never passes that value. No numerical solver of differential equations is used: within each year the equations of all six models have a known exact solution, and that solution is evaluated once per year. The only approximation is the annual time step itself, and re-running the calibrated three-pool model with twelve monthly steps a year changed the national mean stock by at most 0.7 tonnes of carbon per hectare in any year, about one per cent of the stock.

The two ways differ when a pool turns over faster than once a year. The simple way then carries the pool past its balance value in one year and back in the next, and produces an oscillation that belongs to the method of calculation and not to the model. The fastest pool of the three-pool model turns over more than once a year in warm years, so the difference matters here.

All six models use the exact solution. For several pools that pass carbon to one another the exact solution involves the exponential of the matrix of rates and transfers, and the models differ only in how that exponential is obtained. For the one-pool and two-pool models the solution is written out as a formula. For the three-pool model carbon moves in one direction only, from the fast pool to the slow pool and from the slow pool to the humus, so the solution can also be written out as a formula, as a combination of one exponential decay per pool. In the three versions of Yasso carbon moves between pools in every direction, and the exponential is computed numerically in the Fortran code of the models by the standard method of scaling and squaring (Moler and Van Loan, 2003).

2.10 Mean transit time as a common measure of a model

The six models differ in how many carbon pools they contain and in how carbon moves between those pools, so their parameters cannot be compared with one another directly. A single number can be compared, however, and that number is the mean time a unit of carbon spends in the soil between entering it as litter and leaving it as respired carbon dioxide. Sierra et al. (2017) call that quantity the *mean transit time*, and we use their term throughout. Much of the soil carbon literature, including earlier work of our own, calls the same quantity the mean residence time; Sierra and coauthors show that the term residence time has been used for at least three different quantities and discourage its use, so we adopted their convention. At steady state the mean transit time is also equal to the total carbon stock divided by the total input flux, which is what many authors call the turnover time.

For a system in which each pool loses carbon in proportion to the amount of carbon it holds, and in which the rates do not change with time, the mean transit time can be computed from the rates and transfers of the model alone (Rasmussen et al., 2016; Sierra et al., 2017): at steady state the mean transit time equals the total carbon the system holds divided by the flux of carbon entering it. We therefore give each model a litter input of exactly one tonne of carbon per hectare per year and compute the carbon stock at which the gains and losses of every pool balance. The balance stock is computed directly and not by running the model until it settles: for the three simple models the balance stock is written out as a formula, and for the three versions of Yasso the balance equations of all pools are solved together as one linear system. The mean transit time is then the balance stock divided by the input. Because the models are linear, the result does not depend on the amount of

input chosen: doubling the input doubles the stock and leaves the ratio unchanged. An input of one tonne only makes the stock and the mean transit time the same number; the quantity reported is the ratio, in years.

Two properties make the quantity useful as a basis for comparing models. First, the mean transit time computed in this way is a property of each model alone. Neither the litter input multiplier, nor the estimated starting state, nor any soil carbon observation enters the calculation, so two models can be compared even though they were fitted to the same data with different numbers of parameters, and our calibrated parameters can be compared with the published parameters of the same model on exactly the same footing. For the same reason, the mean transit time is not the ratio of simulated stock to simulated input at any date of the simulation, which reflects the imbalance of the soil as well as the model. Second, the calculation collapses a parameter vector of between six and twenty-six numbers into one number with a physical meaning and a unit of years, which allows models that are otherwise very different to be compared on the same axis.

Because a decomposition model responds to climate and to litter chemistry, the mean transit time is not a constant of the model but a property of the model evaluated under a specific condition for these variables. We therefore fix one reference condition and use the same reference condition. The reference climate is the mean climate of the study plots, with a mean annual temperature of 3.4 degrees Celsius, a temperature amplitude of 13.8 degrees and an annual precipitation of 591 millimetres. The reference litter is the mean litter composition of the study plots, normalised to a total of one, which divides the input into 0.766 non-woody litter, 0.219 fine woody litter and 0.014 coarse woody litter, each with its own chemical composition, so that the part of each model describing the effect of the size of woody litter is exercised as it would be in a real simulation. The mean transit times reported here should therefore be read as the mean transit time of each model under such conditions, not as a universal constant of the model, nor as the transit time of the real soil under changing conditions. The comparison between our calibration and the published parameterisation does not depend on the choice of reference, because both sides of the comparison are evaluated at the same reference.

2.11 The error model

Observations are compared with predictions on the logarithmic scale, so that the error is proportional to the size of the stock. The total spread of the error is fixed at 0.800 on the logarithmic scale, about twice the spread of the observed stocks around the national mean of each campaign, 0.39. That spread is the error of an idealised model that reproduces exactly the change of the national mean from one campaign to the next and knows nothing about individual plots. Such a model would follow the measurements together with their errors, because the measurements are all the information it has. A process-based model adds information that the measurements do not contain, in the litter input, the climate and the representation of decomposition, and where the measurements are in error a process-based model is expected to depart from them. The error term has to leave room for that departure.

Four further reasons place the realistic error above that of the idealised model. The litter input carries errors of its own at each plot: the variation of the litter input between plots, 0.41, is followed only weakly by the observed stocks, and counted as an independent error it raises the total to 0.57. The observed errors have heavier tails than a normal distribution. The observed sample excludes outlying plots. Part of the measurement error, in the stone volume, in the bulk density and in the

modelled deep tail of the profile, is shared between plots and does not appear in a spread between plots. A value at the upper end of the plausible range was preferred, because a larger error weakens the pull of the observations relative to the priors, so that results attributed to the observations are stated conservatively.

Observations of soil carbon are not independent of one another. Observations of the same plot share whatever the models get systematically wrong about that plot, observations from the same part of the country share whatever the models get wrong about that part of the country, and observations from the same campaign share whatever the protocol of that campaign did differently. The error of each observation is therefore divided into four parts: a part shared by all observations in the same band of latitude, a part shared by all observations of the same plot, a part shared by all observations of the same campaign, and a part that belongs to the observation alone,

$$\log y_{ij} \;=\; \log f_{ij}(\theta) \;+\; u^R_{r(i)} \;+\; u^P_i \;+\; u^C_j \;+\; e_{ij},$$

where y_{ij} is the stock observed on plot i in campaign j, $f_{ij}(\theta)$ the stock predicted by the model, and u^R, u^P, u^C and e the four parts. Each part is treated as a random error with a mean of zero and a fixed standard deviation, and no part is estimated for any individual plot, band or campaign. Instead, the likelihood is averaged over all the values each shared part could take, weighted by how probable each value is, which for normal errors can be done exactly; the shared parts then disappear from the calculation and leave only a correlation between observations that share a plot, a band or a campaign.

The standard deviations of the shared parts were taken from the observations themselves, after removing the mean of each campaign. The standard deviation of the regional part, 0.117, is the spread between the means of eight bands of latitude, whose boundaries were placed so that each band contains the same number of plots, because the plots are unevenly distributed from south to north and bands of equal width would leave some bands almost empty. The standard deviation of the plot part, 0.396, is measured from how strongly two observations of the same plot vary together. Independent errors add by their variances, that is by the squares of their standard deviations, and not by the standard deviations themselves, so the variances of the four parts add up to the variance of the total, 0.800^2, and the standard deviation of the part that belongs to each observation alone, 0.685, is what remains once the three shared parts are accounted for. Dividing the error in this way does not change how far a single observation may lie from the model, but it changes how much the observations are worth together: observations that share part of their error partly repeat one another, and the 1205 observations carry about as much information on the national mean as 220 independent observations would. Both published calibrations of Yasso assumed the errors of the observations to be independent (Tuomi et al., 2009; Viskari et al., 2022), and the calibration of Yasso20 named the correlation of errors within a time series as an effect still to be included (Viskari et al., 2022). Treating correlated observations as independent leads to parameter uncertainties that are too narrow (Oberpriller et al., 2021), and the shared parts of the error used here are a way of avoiding that.

The standard deviations of the campaign parts cannot be measured from the data, because a shared offset between campaigns and a genuine change of the stock between campaigns look the same in the observations. The standard deviations were therefore prescribed: 0.030 for each of the two later campaigns, which followed similar protocols, and twice that, 0.060, for the first campaign, which differs from the later two in three documented ways. The organic layer of the first campaign

excluded the litter and moss layer, which is added here from each plot's measurement of 2006; the fieldwork of the first campaign extended over a decade; and the carbon content of its mineral layers below 5 cm was mostly predicted from the loss of mass on ignition rather than measured directly (Kramarenko, 2012).

2.12 Sampling

Each model was calibrated with five independent runs of 50 000 iterations, of which the first 5 000 were discarded, using the differential evolution Markov chain sampler of ter Braak and Vrugt (2008), in the implementation provided by the `BayesianTools` package (Hartig et al., 2023), version 0.1.8, under R 4.3.1 (R Core Team, 2023). The sampler proposes each new parameter vector from the difference between past states of its chains, occasionally along the line joining the current state to a past one, so that the proposals adapt to the shape of the posterior; this suits posteriors with strongly correlated parameters, such as the one described in this paper. Each run maintains three interacting chains, so that the five runs together provide fifteen chains. The information that the observations add to each parameter was measured as the Kullback–Leibler divergence of the marginal posterior from the marginal prior, estimated from kernel densities of the samples. The divergence is zero when the posterior is identical to the prior, that is when the observations say nothing about the parameter, and grows as the observations narrow the distribution or move the distribution away from the prior. The divergence is expressed in nats, the unit that results when information is measured with the natural logarithm, as bits are the unit that results with the logarithm to base two. Two cases give a sense of the scale: a posterior with the same centre as its prior and one quarter of its width corresponds to about 0.9 nats, and a posterior with the same width as its prior but moved by one prior standard deviation corresponds to 0.5 nats. Convergence was assessed with the potential scale reduction factor and with the effective sample size, and the rate of proposals rejected as numerically impossible was monitored throughout.

2.13 Predictive evaluation and residual analysis

Predictions were generated for both the calibration plots and the held-out plots by drawing from the posterior and running each draw forward. Model skill is reported as the coefficient of determination and the root mean squared error, separately for the calibration plots and for the held-out plots. Residuals were defined as the logarithm of the observed stock minus the logarithm of the predicted stock, so that a positive residual means that the soil holds more carbon than the model predicts. The residuals of the six models are nearly identical from plot to plot, with a correlation of 0.99 or higher between any two models, and the detailed analyses of the residuals therefore use a single value for each observation, the mean of the residuals of Yasso07 and Yasso15.

Three analyses were made. First, a random forest was fitted to the residuals of each model and evaluated out of bag, with about sixty site variables describing the climate, the texture and chemistry of the soil, the composition of the litter input and the stand in 1985, to measure how much of the residual can be predicted at all and which variables lead. The ranking was checked with a wider set of several hundred variables: the stand records and tree tallies of the four inventories, the chemistry of the soil, topographic wetness, nitrogen deposition, the record of management, and the cover of the understorey vegetation measured in 2023; variables that measure the soil carbon itself were left out. The leading variables of the forest then guided two detailed analyses, made on

the mean residual of Yasso07 and Yasso15. Second, the mean residual of each plot was regressed on the basal area of the stand in 1985, fitted as a smooth curve, together with three properties of the site: the fertility of the site, expressed by the forest site type recorded by the inventory; the texture of the mineral soil, recorded in the field in 1985; and the volume share of stones in the soil. Third, the residual of each soil observation was related to the time elapsed since the last regeneration cut before the soil was sampled, a regeneration cut being a felling that removes the stand so that a new one can be established. The year of the cut was taken from the stand records of the inventories of 1985, 1990, 1995 and 2021–2023 and from the forest use declarations filed with the Finnish Forest Centre from 1997 onwards. Where no cut was recorded before the sampling, the time since the stand was established was used instead, taken as the age of the stand in 1985, measured on the tally trees, plus the years elapsed since 1985. Because each plot contributes up to three observations, the comparison between the basal area of 1985 and the time since the cut used a regression with a separate intercept for each plot.

2.14 Forward simulations

Two side analyses extend the calibrated models beyond the observations, to give an order of magnitude for the accumulation that remains rather than a forecast. Both use the same 100 posterior draws as the predictions and the plots measured in all three campaigns. The projection runs each model for 60 years after the last observation, with the litter input held at its value in the last observed year, 2024, and with the climate of the last 20 observed years repeated three times, so that the projection contains neither a change of climate nor a change of forest growth. Because the repeated climate alternates between a cooler and a warmer decade, rates of change over the projection are computed over whole 20-year cycles. The equilibrium stock of each plot is the stock at which gains and losses would balance if the same litter input and climate were held indefinitely, computed with the routine of each model for the state of balance, as for the mean transit time (Section 2.10); the distance from balance is the equilibrium stock divided by the stock of 2024, minus one. The input scenario repeats the projection with the litter input increased by 20 per cent from the first projected year onwards. Because the models are linear, the eventual gain of stock under that scenario is 20 per cent of the equilibrium stock, and the scenario shows how much of the gain is realised within 60 years.

3 Results and discussion

3.1 The models reproduce the national trajectory, and the fit cannot tell them apart

All six calibrated models reproduce the mean soil carbon stock of the three campaigns to within 3.5 tonnes of carbon per hectare (Figure 4), and over the full observed window the six models bracket the observed rate of change of +0.259 tonnes of carbon per hectare per year, from below in SP1 and Yasso20 and from above in TP2 and TP3. Agreement of this kind is expected, because the three campaigns are the data the models were calibrated on, and it shows consistency with the observations rather than predictive ability (Le Noë et al., 2023). The observations also carry their own uncertainty: the three campaigns followed three protocols and the first campaign spans a decade, so that the rate over the full window survives a shared offset between campaign levels

of about 9 per cent, whereas the rate over the last interval, on which the apparent slowing of the accumulation rests, is removed by an offset of about 2 per cent. Which model lies closest to the observed rate cannot therefore be read as evidence that the model is better.

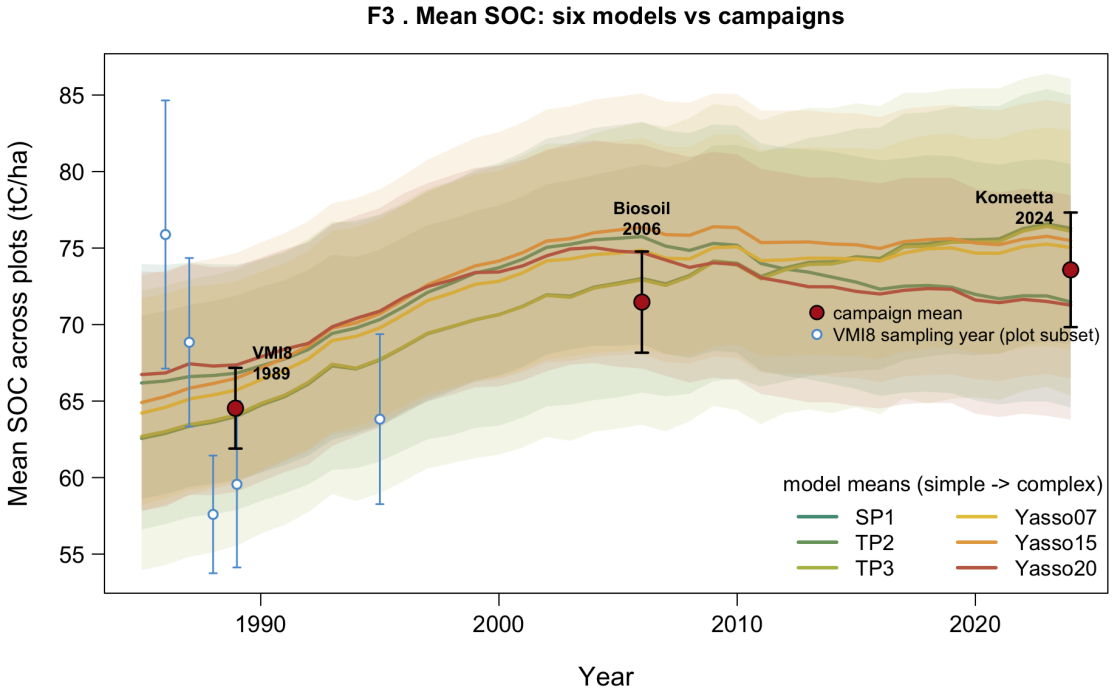


Figure 4: Modelled and measured mean soil carbon, 1985–2024. Lines are the mean stock across plots for each of the six models, and the shaded bands are 95% posterior intervals. Markers are the measured campaign means with their confidence intervals, calculated on the same set of plots as the model curves. All six models pass within 3.5 tonnes of carbon per hectare of every campaign mean, and the shape of the modelled trajectories separates the six models into three groups.

[Supplementary figure: F5_stock_change]

The fit does not separate the model structures either. Skill does not increase with the number of pools: the coefficient of determination is 0.016 to 0.021 on the calibration plots and no more than 0.0012 on the held-out plots, and the root mean squared error on the held-out plots differs by 3.6 per cent between the best and the worst model (Table 1). Skill is measured plot by plot, so the coefficient of determination measures how well the models separate one plot from another, and none of the six represents the local variation that separates them (Section 3.7); the national trajectory, which all six reproduce, is a different quantity. Inside the models, the observations leave most parameters undetermined: all parameters converged, but the transfer fractions stay close to their priors, gain less than one nat of information, and trade off against the other fractions leaving the same pool. The observations determine the national level and trend of the soil carbon stock, and not how each model produces them.

[Supplementary figure: F6b_skill_vs_complexity] [Supplementary figure: F8b_fraction_ridges]

The six trajectories nevertheless differ in shape, and the shape is what an extrapolation rests on. Yasso07 and Yasso15 rise and then flatten, TP2 and TP3 keep rising, and SP1 and Yasso20 reach a maximum in the late 2000s and then decline. Models that agree within the observed window can

Model	Pools	Free params	R^2 (cal)	R^2 (hold)	RMSE (hold)	Eff. flux $\sigma_{\text{inp}}\bar{J}$	σ_{init}
SP1	1	6	0.017	0.000	47	3.2	0.84
TP2	2	7	0.016	0.000	48	4.3	0.50
TP3	3	9	0.015	0.000	49	4.3	0.49
Yasso07	5	20	0.019	0.000	48	4.0	0.52
Yasso15	5	26	0.021	0.001	47	3.9	0.54
Yasso20	5	26	0.016	0.000	47	4.2	0.59

Table 1: Summary of the six calibrated models. Number of carbon pools, number of free parameters, coefficient of determination on calibration plots and on held-out plots, root mean squared error, the effective litter flux implied by the calibration, and the estimated ratio of the litter input of 1917 to the litter input of 1985. Skill does not increase as pools are added.

therefore disagree outside it.

3.2 The calibration settles at a larger litter input, and in two versions of Yasso at a faster turnover, than the published parameterisations

The calibrated models sit away from their published parameterisations in two quantities at once, and the displacement in transit time is graded across the three versions of Yasso rather than uniform. The effective litter input, defined as the multiplier σ_{input} times the litter input carried by the product, ranges from 3.19 to 4.34 tonnes of carbon per hectare per year, against the 2.511 tonnes that the product supplies. The intrinsic mean transit time is 24.6 years for Yasso07, 27.1 years for Yasso15 and 22.6 years for Yasso20, against published values of 33.5, 30.4 and 21.9 years respectively. Expressed as a ratio of the calibrated transit time to the published transit time, the three versions of Yasso give 0.73, 0.89 and 1.03.

Yasso20 therefore serves as a negative control, and the negative control is worth stating explicitly. The calibration displaces the transit time of Yasso07 strongly, displaces the transit time of Yasso15 moderately, and does not displace the transit time of Yasso20 at all: the calibrated value for Yasso20 lies three per cent above the published value, and the 90 per cent posterior interval of the calibrated value, from 19.6 to 26.4 years, contains the published value comfortably. A calibration that moved every model in the same direction by a similar amount would be open to the objection that the movement came from the data treatment or from the likelihood rather than from the models. A calibration that moves two models and leaves the third where its authors put it is not open to that objection.

All transit times quoted here are compared against the published *point* parameterisation. Yasso15 and Yasso20 also publish posterior samples, and because transit time is a non-linear function of many parameters, the median transit time of a published sample is not the transit time of the median published parameters. The two comparators are never mixed in this paper.

The displacement in litter input and the displacement in transit time are one displacement and not two. The observed stock constrains the product of the litter input and the transit time, so a soil that receives more litter and decomposes it faster fits the observations as well as a soil that receives less litter and decomposes it more slowly. The calibration moves along that trade-off. Moving along the trade-off back towards the published transit time costs little in fit. Among the posterior draws that lie within five per cent of the published transit time, the best fit is at most 9.7 log-

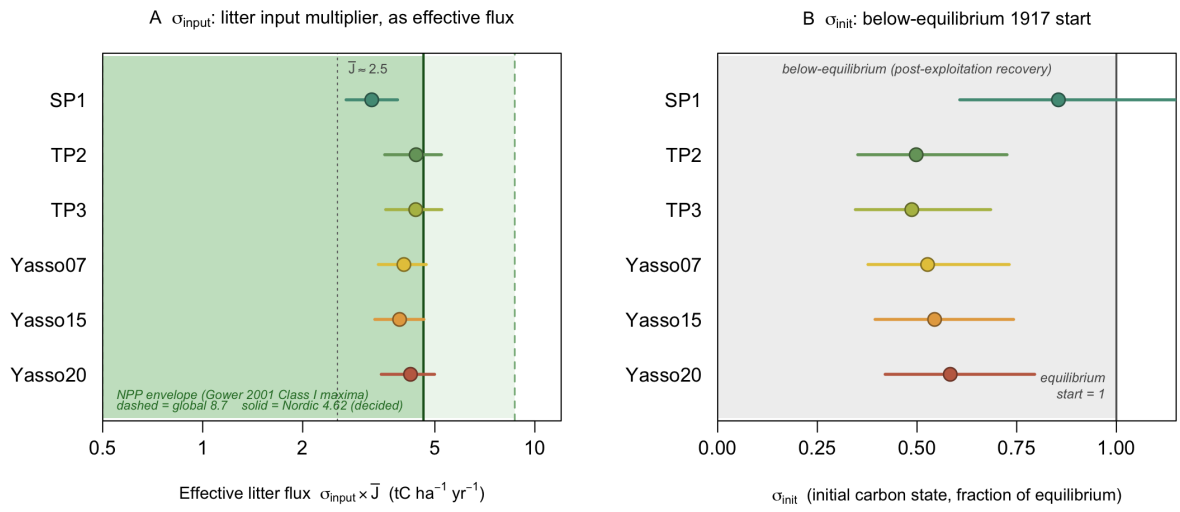


Figure 5: The two estimated auxiliary quantities, read as physical quantities. *Left:* the litter input multiplier expressed as an effective flux of litter carbon to the soil, that is, the multiplier times the litter input that the product supplies. The dotted line marks the flux the product itself supplies, and the shaded bands mark the envelope of independent estimates of boreal net primary production. Every model settles above the tree litter the product supplies and inside the envelope. *Right:* the estimated ratio of the 1917 litter input to the 1985 litter input. All six models place the 1917 input below the 1985 input, which is the direction the forest history requires.

likelihood units below the best fit overall for Yasso07, at most 3.0 units for Yasso15, and no lower for Yasso20, and 6.5 per cent of the posterior of Yasso15 lies at or beyond the published value. The soil measurements therefore suggest a direction rather than impose a value.

[Supplementary figure: S13_mrt_ridge_benchmark]

The trade-off is not created by the choice of starting state, and a transient starting state does not remove it. The trade-off follows from the condition of balance itself, so the trade-off exists under either choice of starting state. What a transient starting state adds is information about the rate at which the soil approaches its eventual stock, and the rate of approach depends on the transit time alone. Estimating the starting state therefore narrows the trade-off, and the narrowing is partial: across the three versions of Yasso the transit time and the litter multiplier still correlate between -0.64 and -0.73.

3.3 The displacement in litter input follows the data more than the prior

The litter input multiplier could in principle be an artefact of the prior placed on it. To test that possibility, the calibration was repeated with the centre of that prior moved from 1.08, the value derived from Lehtonen and Heikkinen (2016), to 1.27, the value implied by the reconstruction of Liski et al. (2006), with everything else unchanged. Moving the prior centre by 18 per cent moves the posterior median only about one third as far: from 1.54 to 1.62 in Yasso15, from 1.73 to 1.87 in TP2, and from 1.27 to 1.35 in SP1. Under the higher prior centre the posterior median still sits between 1.28 and 1.50 times above the prior centre, and under the lower prior centre the posterior median sits between 1.17 and 1.60 times above it.

Because the posterior follows the prior centre only partially, and because the posterior remains

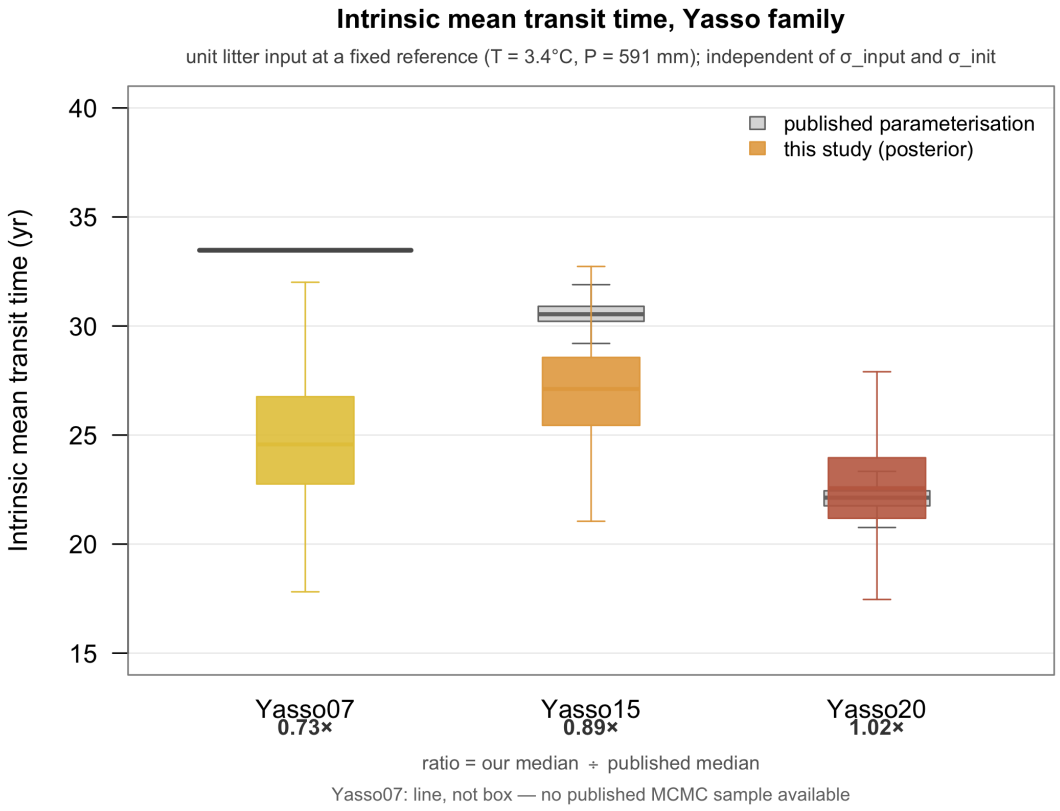


Figure 6: Mean transit time of carbon in the three versions of Yasso, calibrated and published. Transit times are computed at a common reference climate and litter composition, from the model equations alone, so that the comparison depends neither on the litter input multiplier nor on the starting state.

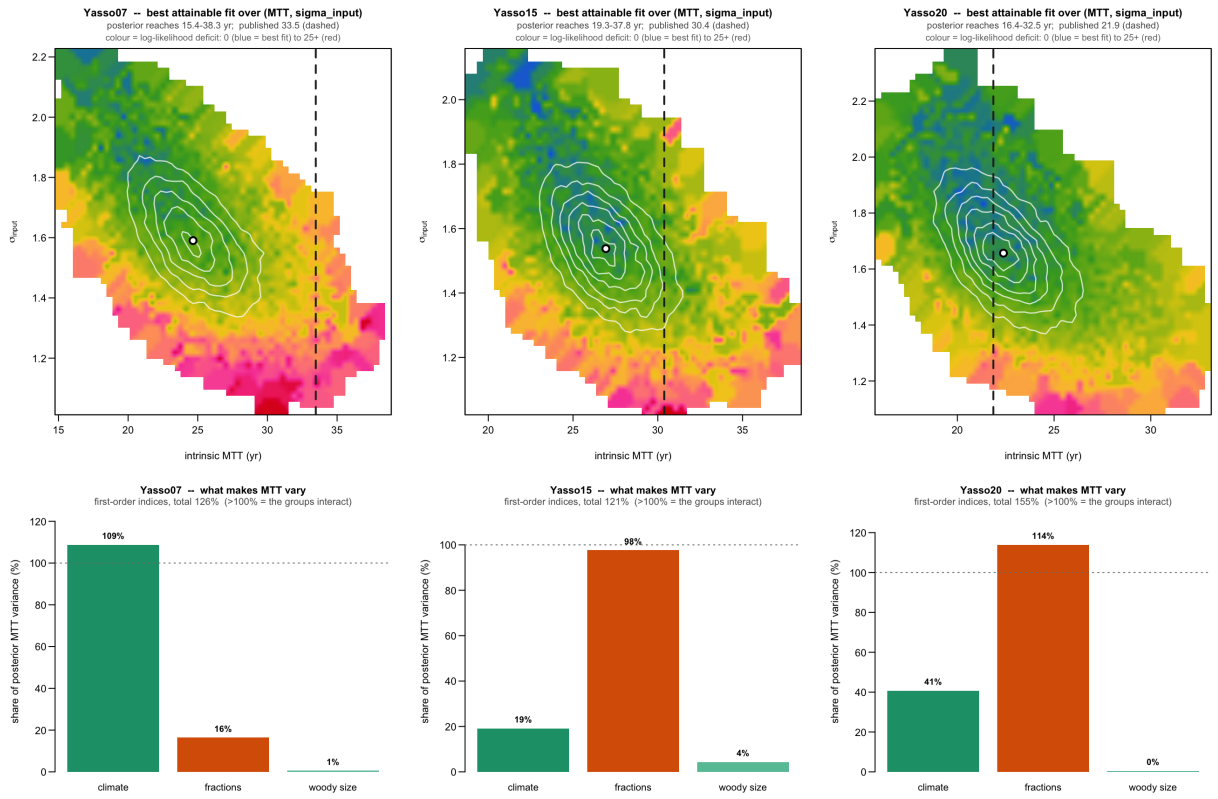


Figure 7: The trade-off between transit time and litter input. For each version of Yasso, the best fit to the data attainable at each combination of mean transit time and litter input multiplier. **The colours show the fit to the data alone, with the priors excluded**, from blue at the best fit through green and yellow to red at the worst. **The white contour lines show the posterior, which combines the fit to the data with the priors.** The dashed line marks the published transit time and the white dot marks the posterior median. Cells that the sampler visited fewer than five times are left blank, and a blank cell means that the region was not visited rather than that the region cannot be fitted. The offset between the bluest colour and the innermost contour measures how far the priors pull the answer away from where the data alone would place it.

well above the prior centre under either choice, the preference for a larger litter input comes mainly from the soil measurements rather than from the prior. The preference survives either anchor, and choosing between the two anchors is therefore a question about how the prior should be documented and not a question about whether the result holds.

3.4 What might be missing from the litter input

The multiplier σ_{input} is worth reading as a physical quantity and not as a nuisance parameter, because the multiplier is not scale-free. Multiplied by the litter input the product supplies, the multiplier becomes an effective flux of litter carbon to the soil, and an effective flux can be compared with independent budgets. The gap to be explained is between 0.68 and 1.83 tonnes of carbon per hectare per year.

The most likely explanation is not that the litter model is wrong but that the litter model is incomplete. Litter products of this kind are trusted because they have worked for decades, and one reason they have worked is that a calibration free to adjust its decomposition rates can absorb a missing flux without any visible misfit. A calibration performed on externally anchored decomposition rates removes that compensation, and a missing flux then has to appear somewhere.

3.4.1 Carbon released by living roots

Carbon released into the soil by living roots belongs to no biomass compartment, so a litter model assembled compartment by compartment cannot represent it. Agricultural carbon accounting has treated the same flux as a standard term for two decades. The widely used approach of Bolinder et al. (2007) adds an explicit pool of extra-root material set at 65 per cent of the standing root biomass, derived as the ratio of the share of below-ground carbon released by living roots and retained in the soil to the share that remains in the roots themselves. The extra-root pool is defined as "exudates, as well as root hairs and fine roots sloughed off during the growing season which, because of sampling difficulties, are not included in the 'root' fraction" (Bolinder et al., 2007). Nothing of the kind appears in the forest inventory chain.

Carbon released by living roots also forms stable soil organic matter more efficiently than above-ground litter does. In the Ultuna long-term experiment the humification coefficient fitted for root-derived carbon was about 2.3 times the coefficient fitted for the same mass of above-ground crop residues (Kätterer et al., 2011). A below-ground input that is both unaccounted for and disproportionately efficient at forming stable organic matter is a candidate of the right kind for a displacement inferred from stocks.

The agricultural coefficient cannot be transferred to forests unchanged. In an annual crop the root biomass is measured once, at harvest, so the coefficient of 65 per cent compensates for everything the single measurement misses, including the roots that grew and died during the season. A forest litter chain already represents the turnover of fine roots explicitly, through a turnover rate applied each year, so the forest equivalent of the extra-root term covers less. The forest equivalent covers more than nothing, however, because forest fine-root biomass is itself estimated from soil cores, and soil cores miss sloughed material for exactly the reason Bolinder et al. (2007) give.

3.4.2 Understorey vegetation

The litter product used here represents tree litter and does not represent understorey vegetation, by design. The national inventory does represent understorey vegetation, and values it at 0.506 tonnes of carbon per hectare per year in southern Finland and 0.666 in northern Finland (Table 6.4-3, after Statistics Finland, 2024; Muukkonen and Mäkipää, 2006). The same values appear in the appendix of Lehtonen and Heikkinen (2016). Understorey vegetation is therefore both the best-quantified of the candidate fluxes and a flux that the input used here is known to omit.

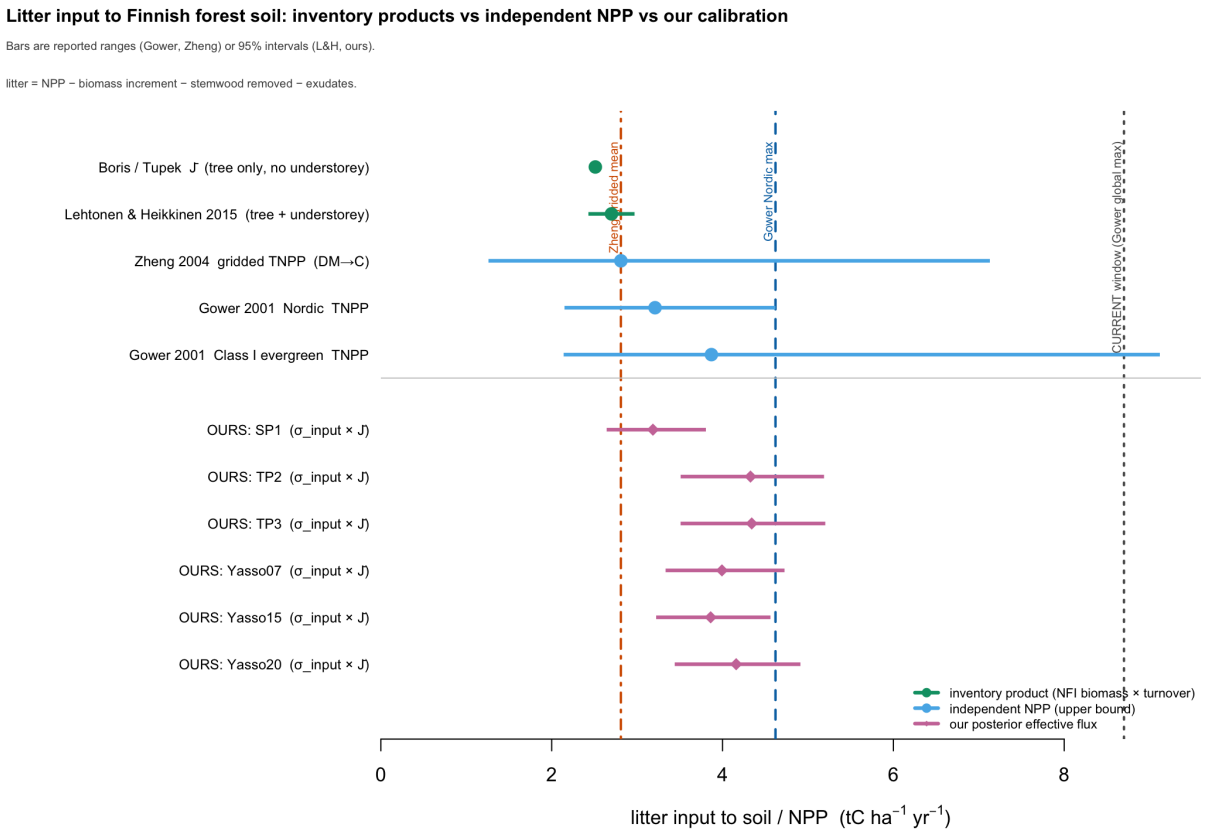


Figure 8: The litter input inferred by the calibration, against independent estimates of the same flux. Green symbols are inventory-based litter products, blue symbols are independent estimates of net primary production, which bound the litter flux from above, and the lower group shows the effective flux inferred here for each of the six models. Bars are reported ranges for the literature estimates and 95% posterior intervals for the models.

The share of the understorey rises towards the north, so a single national multiplier applied to a tree litter product must be wrong in opposite directions at the two ends of the country. Lehtonen et al. (2016) reached the same conclusion from the other direction, reporting that "the role of understorey litter input was underestimated when the Yasso07 model was parameterised, especially in northern Finland".

3.4.3 Mycorrhizal fungal residues

Where boreal soil carbon comes from has been revised substantially over the last decade. Using bomb radiocarbon across a chronosequence of forested islands, Clemmensen et al. (2013) attributed 50 to 70 per cent of the stored soil carbon to roots and to root-associated organisms rather than to above-ground litter, and identified mycorrhizal mycelium as a central agent of the storage. Production of mycelium in coniferous stands is measured in hundreds of kilograms per hectare per year (Ekblad et al., 2013; Hagenbo et al., 2017), of which only a part becomes input to the soil.

Mycelium is not a tree biomass compartment, so no turnover coefficient in a litter chain can produce it, and the flux is missing by construction rather than underestimated. The figure of 50 to 70 per cent should not be read as the size of the missing flux, because the figure describes the origin of carbon that has been stored and part of that carbon already enters our input as fine-root litter. What can be said is that the dominant precursor of stable boreal soil carbon is root-derived and partly fungal, and that the fungal part is outside inventory litter models.

3.4.4 Fine root lifespan, where the evidence points the other way

Fine roots are the largest below-ground litter term, and fine-root litter enters the inventory chain as a biomass multiplied by a single turnover rate of 0.85 per year applied across species and regions. The lifespan behind that rate is among the least settled quantities in boreal ecology. Direct observation with minirhizotrons puts the median lifespan of Norway spruce fine roots below two and a half years, whereas radiocarbon dating of the same species returns apparent ages of up to fourteen years (Strand et al., 2008).

A reconciled mean lifespan of about 3.8 years would imply a turnover rate near 0.26 per year, which is roughly a third of the rate in use, and a lower turnover rate would make fine-root litter smaller rather than larger. The fine-root evidence therefore points in the opposite direction to the other three candidates, and would widen rather than close the gap. We report the fine-root evidence as it stands. A section in which every candidate mechanism happened to support the conclusion would be less credible than one in which a candidate does not.

3.4.5 Early-succession vegetation

Litter models keyed to standing tree biomass are least informative in the years immediately after a stand-replacing harvest, because tree biomass is then near zero while the site is neither bare nor unproductive. Ground vegetation biomass falls sharply at clear-cutting and then recovers through a pioneer phase in which a large share of annual production returns to the soil each year (Palviainen et al., 2005). Whether the total over a rotation exceeds what a stand-keyed model assumes is not established, and we raise early-succession vegetation as a gap in the timing of the input rather than as a quantified addition to its total.

3.4.6 What the candidates add up to, and what they do not

Two of the candidate fluxes can be given Finnish numbers. Understorey vegetation supplies between 0.51 and 0.67 tonnes of carbon per hectare per year, and mycorrhizal residues plausibly supply a further 0.2 to 0.5. The two together reach the lower end of the gap of 0.68 to 1.83. Carbon released by living roots, priced at the agricultural coefficient and applied to the fine-root share of our own

litter product, would add a further 0.73, which taken at face value would carry the total past the upper end of the gap.

The candidate fluxes should therefore be read as a set that brackets the gap and not as a sum that matches it. A budget that added up exactly would be the more suspicious result, because the candidate fluxes overlap, because the agricultural coefficient must be smaller in forests, and because one of the candidates points the other way. The calibration settles at a larger litter input than the product supplies, several physically plausible sources of the difference exist, and identifying which of them is responsible is not a task this study can complete.

3.5 Does it matter where the calibration lands on the trade-off?

The soil measurements cannot choose a point on the trade-off between litter input and transit time, so the choice has to be judged by its consequences. If moving along the trade-off changes nothing that can be observed, then the displacement reported above is bookkeeping. If moving along the trade-off changes the projection, then the displacement matters to anyone who uses these models to project.

We compared the published parameterisation and our calibration by running both forward under the same warming scenarios, without recalibrating either, and with only the litter input multiplier refitted so that both reach the observed window equally well. Because both arms reproduce the observed window, any divergence after the observed window is extrapolation and not fit.

Only Yasso07 separates the two arms. In Yasso07 the two arms differ by -2.08 tonnes of carbon per hectare, with an interval from -3.57 to -0.79, under the SSP2-4.5 scenario. Yasso15 and Yasso20 do not separate at all.

The reason only Yasso07 separates is structural. Yasso07 applies a single climate modifier to every carbon pool, so the turnover of the soil and the sensitivity of the soil to temperature are governed by the same parameters, and moving along the trade-off moves both together. Yasso15 and Yasso20 apply separate climate modifiers to different pools, and the pools that hold most of the carbon respond almost identically in the two arms. [Supplementary figure: A1_forward_appendix]

The result about Yasso15 and Yasso20 is not that they are right. The result is that Yasso15 and Yasso20 are insensitive to a calibration choice that the available data cannot settle, which is a narrower statement than saying that our calibration changes the projection, and a more useful one for an inventory.

3.6 Holding or increasing the litter input

With the litter input held at its present value, five of the six models place the soil well below its state of balance: the equilibrium stock lies 26 to 42 per cent above the stock of 2024 in TP2, TP3 and the three versions of Yasso (27 per cent in Yasso07, 42 in Yasso15, 35 in Yasso20). SP1, whose single pool turns over slowly and uniformly, is within one per cent of balance. Liski et al. (2006) estimated that Finnish upland soils would eventually settle at a level 38 per cent above that of 1922, a value inside the range of the three versions of Yasso. The two estimates were obtained by different routes, a reconstruction started in balance and a calibration started out of balance, and agree in magnitude although their reference years differ.

The distance from balance is realised slowly. Over the first 20 years of the projection the sink is 0.07 to 0.11 tonnes of carbon per hectare per year in the three versions of Yasso and 0.21 to 0.22 in

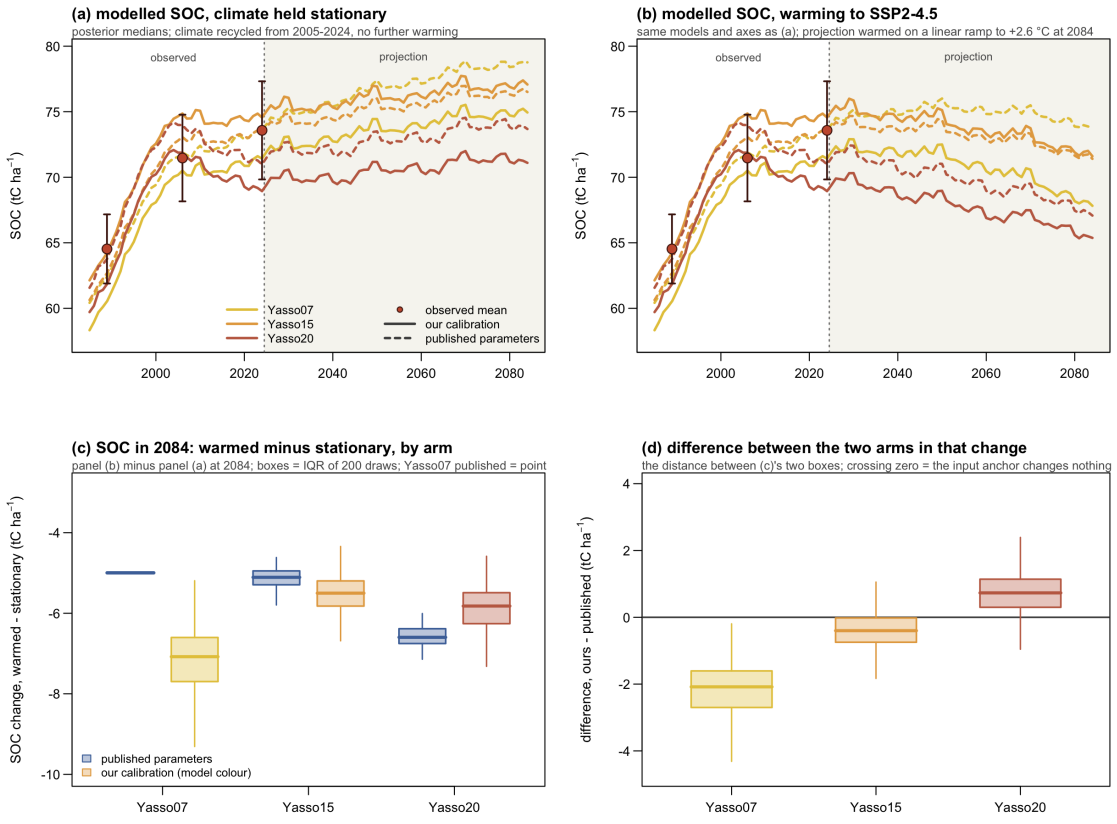


Figure 9: Projections from the published parameterisation and from the calibration, under the same scenarios. Neither arm was recalibrated, and only the litter input multiplier was refitted, so that both arms reproduce the observed window equally well and any later divergence is extrapolation rather than fit. Only Yasso07 separates the two arms.

TP2 and TP3; over the last 20 years it has fallen to about 0.05 in the versions of Yasso and 0.13 in TP2 and TP3, and by 2084 the versions of Yasso have realised 14 to 21 per cent of their distance from balance. None of the six models turns into a source. The sink declines within decades because the pools that turn over fastest approach balance first, whereas most of the remaining distance lies in the slowest pool, which approaches balance over centuries (Sierra et al., 2018). Under a constant litter input, the sink therefore saturates long before the stock does.

The remaining sink is inherited from the estimated starting state. Within each model, the parameter σ_{init} correlates with the distance from balance at -0.79 to -0.97 and with the projected sink of the first 20 years at -0.89 to -0.95 across the posterior draws: the lower the litter input of 1917 relative to that of 1985, the more the soil still has to gain. The calibration places σ_{init} at 0.49 to 0.60 in five of the six models, with upper 90 per cent limits of 0.64 to 0.76, whereas the growing-stock record of the national forest inventory implies about 0.90 and the reconstruction of Liski et al. (2006) about 0.86; only SP1, at 0.83, is compatible with that record. In absolute terms the calibrated litter input of 1917 is not low, between 0.83 and 0.97 times the input the product supplies for 1985 in the five models, because σ_{input} raises it; what departs from the record is the rise of the input between 1917 and 1985, a factor of 1.7 to 2.1 in the calibration against about 1.1 in the growing stock. Of the two halves of the starting state, the ratio of inputs rests on a growing stock that the forest inventory has measured since the 1920s, whereas the balance of the soil of 1917 with its own input rests on no measurement at all, and the centuries of slash-and-burn cultivation, tar burning and forest grazing described in the Introduction argue against it. A soil depleted by that history would lie below balance with its own input, and a single parameter can express such a deficit only as a lower input. We therefore read σ_{init} as an effective starting deficit of the soil rather than as a ratio of litter inputs. On that reading, the soil of 1917 was more depleted than its litter input alone would imply, the remaining sink is conditional on that depletion, and a soil assumed to start in balance with its present input would have almost nothing left to gain under a constant input, as SP1 illustrates.

Increasing the litter input by 20 per cent from 2025 raises the stock by 6 to 10 tonnes of carbon per hectare by 2050 and by 9 to 13 tonnes by 2084, relative to the projection with a constant input, which corresponds to an additional sink of 0.2 to 0.4 tonnes of carbon per hectare per year averaged over 2025–2050. Within 60 years the versions of Yasso realise about half of the eventual gain, 46 to 55 per cent. The gain is larger than the sink of the projection with a constant input, but the gain is finite, largest in the first decades, and lasts only as long as the larger input is maintained. Neither projection contains a change of climate, and neither is a forecast.

3.7 The models reproduce the national mean but not the differences between plots

The same models that reproduce the national mean trajectory have no skill in distinguishing one plot from another, and the two results are not in contradiction. The national mean depends on the total input of litter and on the rate at which the soil turns that input over, which are the two quantities the calibration constrains. The difference between two plots depends in addition on local ecological variation, in the history of each plot and in the properties of its soil, and on how accurately the litter input of each plot is known, and none of the six models is given that information: all plots share one estimated initial state, no model has a term for the properties of the soil, and the litter input of each plot comes from a product keyed to its standing trees. Errors of that kind largely

cancel in a mean over several hundred plots and remain in full on each individual plot. The national inventory reports national and regional totals, for which the first result is the relevant one, and what the models miss between plots is not noise: as shown in Section 3.8, about half of it can be recovered from site variables that are already available.

Plot by plot, none of the six models has any skill. The coefficient of determination on held-out plots is zero to three decimal places in all six models. The informative part of the comparison is not the absence of skill but the structure of what the models predict. The spread of the predicted stocks is wider than the spread of the observed stocks, and the predicted stocks are ordered by the standing basal area of the plot, whereas the observed stocks are not.

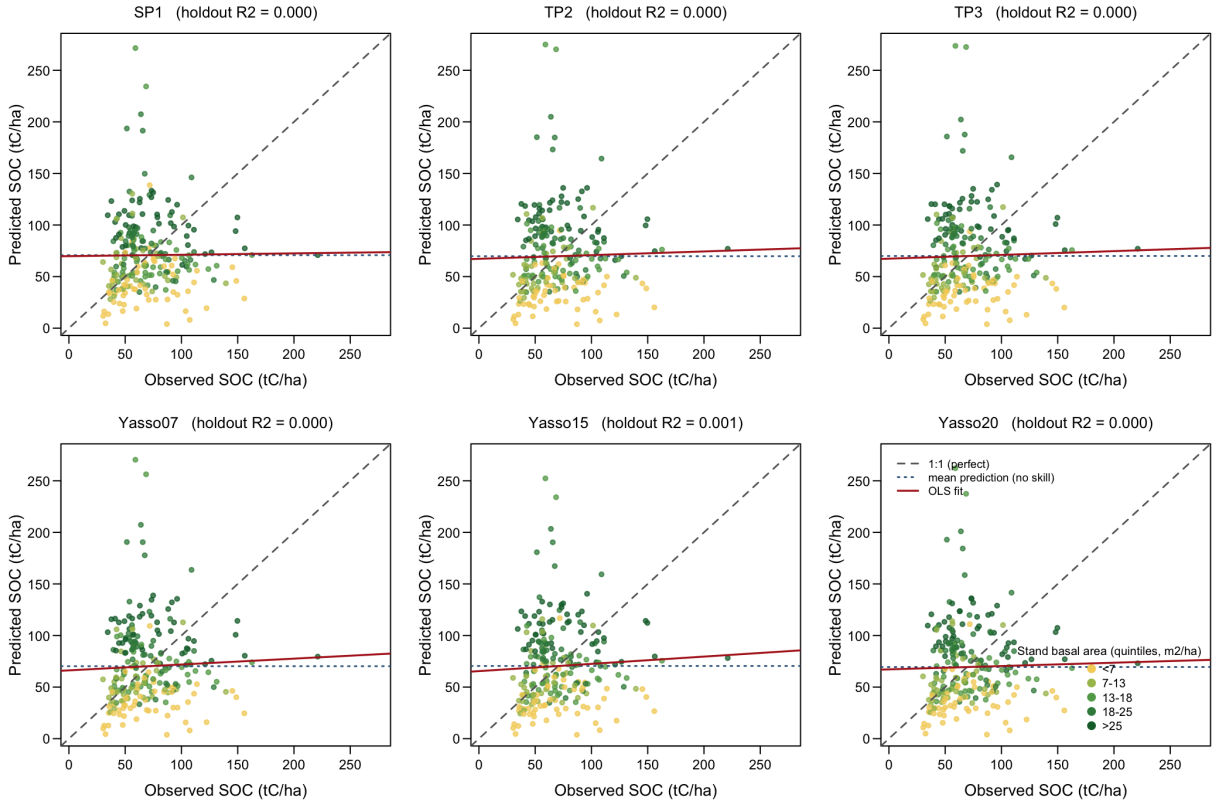


Figure 10: Predicted against measured soil carbon on held-out plots. Points are coloured by the standing basal area of the plot, in quintiles. The dashed line is the one-to-one line, the dotted line is the prediction of the mean, and the solid line is the fitted relation. The predicted stocks spread more widely than the measured stocks and are ordered by basal area, whereas the measured stocks are not.

Measured directly, the slope of the logarithm of the stock against basal area is between +0.044 and +0.049 per square metre per hectare in the six models, and +0.0044 in the observations. The residual therefore falls with basal area in all six models, with slopes between −0.032 and −0.044 and with p values from 10^{-16} to 10^{-111}, on the calibration plots, on the held-out plots, and on the two pooled. Adjusting the observed relation for region and for soil type reduces the observed slope further, to +0.0018, at which point the observed relation is no longer statistically significant.

The gradient does not originate in the models. The six models contain no term in basal area at all. What the six models contain is the litter input, and the litter input itself rises with basal area at a slope of +0.072, which is a span of about eighteen-fold across the observed range of basal

area. The modelled stocks rise with basal area at a slope of +0.073 to +0.076. The models therefore transmit the gradient of the litter product essentially unchanged, and the forward simulation then damps the gradient rather than amplifying it.

We tested whether the discrepancy could instead be a mismatch between compartments. The stock simulated by Yasso covers dead organic matter as well as soil organic matter, and woody litter enters the simulated stock, whereas the observations are soil measurements that cannot contain a stump. Setting the woody litter inputs to zero moves the modelled gradient by six per cent, from 0.0755 to 0.0710 in Yasso15, against a fifteen-fold difference to the observed gradient. The three simple models provide the decisive control, because the three simple models carry a single total litter flux with no information about composition at all, and the three simple models produce the same gradient. A mechanism that is absent from three of the six models cannot explain a gradient that is present in all six.

Because the gradient is transmitted rather than generated, the discrepancy is a statement about the litter product and about the soil, and not about the decomposition models. The litter product states that the carbon input of a plot varies about eighteen-fold across the range of basal area. The soil states that the carbon stock of a plot varies by a factor of about 1.2 across the same range. Two readings follow, and the two readings are not exclusive.

The first reading is that the carbon stock of an individual plot is not set by the current litter input of that plot. Each plot sits at a different point on its own history of disturbance, and one initial state estimated for the country as a whole cannot express what makes individual plots differ from one another. The first reading is the argument of this paper applied one level down.

The second reading is that an eighteen-fold gradient in litter input between plots is not credible. A litter product keyed to standing tree biomass gives a freshly cut plot almost no input, and a freshly cut plot does not stop receiving carbon: dying roots, harvest residues, the ground vegetation and the pioneer trees of the next stand continue to supply it. Section 3.8 shows that the two readings meet in the same plots, and that the history of each stand carries most of what the models miss between plots.

3.8 What the models leave behind follows the history of each stand

About half of what the models leave behind can be predicted from site variables that are already available. A random forest fitted to the residuals of each model explains between 44 and 65 per cent of the residual variance, evaluated out of bag, against a repeatability of the observations of about 0.49 (Figure 11). Because the recoverable share is larger than the noise of the observations, what the models leave behind is not a ceiling set by measurement error.

The same variables lead in all six models, and one variable dominates: the basal area of the stand in 1985. The development class of the stand in 1985 follows, and so does the content of coarse fragments, a physical property of the soil; the importance of every other variable is small. The detailed analysis below therefore starts from basal area and follows the two leads that the forest gives: the properties of the soil, and the stage of development of the stand. A wider set of several hundred variables, drawn from the stand records and tree tallies of all four inventories, the chemistry of the soil, the terrain and the record of management, predicts the residual no better (56 per cent out of bag) and confirms the ranking. Almost all of the recoverable share is carried by four variables: the basal area of the stand in 1985 and three properties of the site, which together explain 53 per cent of the residual variance (Figure 12).

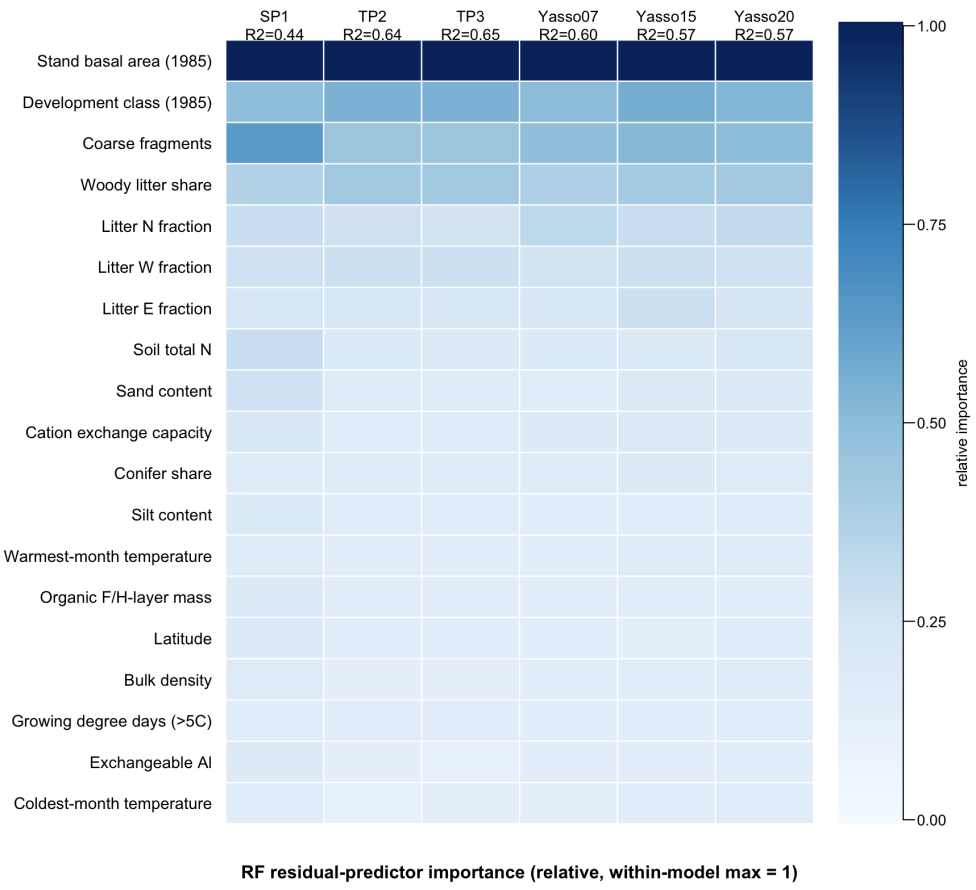


Figure 11: Site variables that predict what the models leave behind. Relative importance of each site variable in a random forest fitted to the residuals of each model, scaled within each model so that the most important variable equals one; the out-of-bag coefficient of determination of each forest is given above its column. The same variables lead in all six models: the basal area of the stand in 1985 first, then the development class of the stand in 1985 and the content of coarse fragments in the soil.

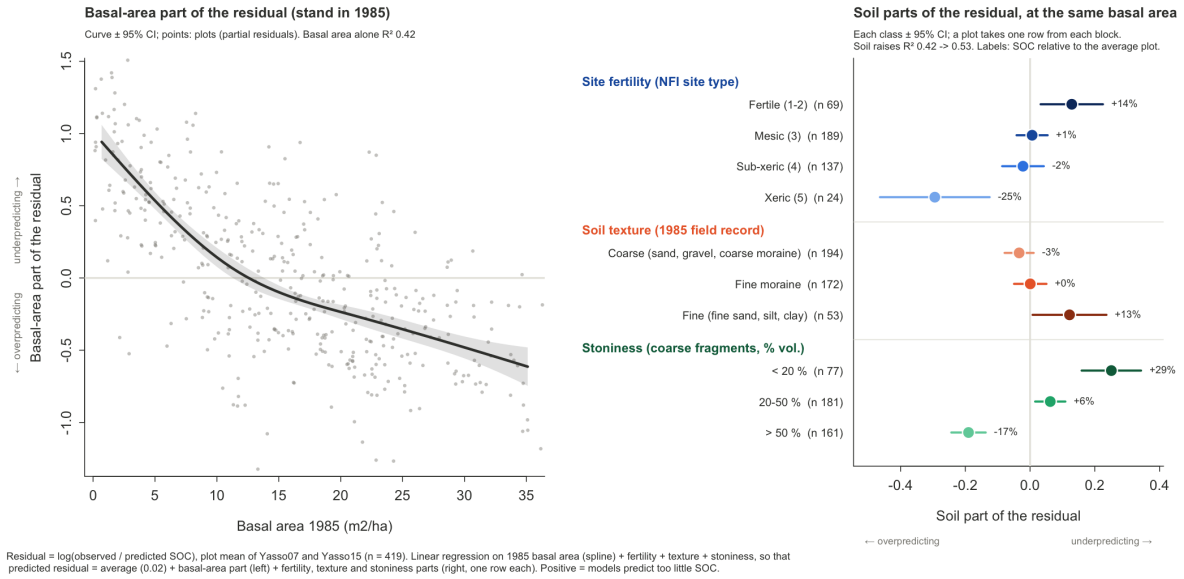


Figure 12: What the models leave behind, in two parts: the stand of 1985 and the soil. The residual is the logarithm of the measured over the predicted soil carbon, as the mean of Yasso07 and Yasso15 over the observations of each plot (419 plots); a positive residual means that the soil holds more carbon than the models predict. One regression on the basal area of the stand in 1985 (a smooth curve) and on the fertility, texture and stone content of the site divides the predicted residual of each plot into parts, each measured from the average plot. Left: the part due to basal area, with its 95% confidence band; the points are the plots. Right: the part due to each property of the site, with 95% confidence intervals, at the same basal area; the labels translate each part into soil carbon relative to the average plot. Basal area alone explains 42 per cent of the residual variance, and the three properties of the site raise the share to 53 per cent.

The basal area of the stand in 1985 is the dominant part. The residual falls steeply with basal area: in the sparsest stands the soil holds about two and a half times the carbon that the models predict, and in the densest stands about half of it, measured from the average plot. Basal area alone explains 42 per cent of the residual variance. Section 3.7 showed where the gradient comes from: the litter product passes the basal area of each stand into its litter input, and the soil does not follow.

The properties of the site add a smaller part, and each part points in an ecologically sensible direction. At the same basal area, fertile sites hold about 14 per cent more carbon than the average plot and the poorest, xeric sites about 25 per cent less; fine-textured soils hold about 13 per cent more; and soils with little stone hold about 29 per cent more, while very stony soils hold about 17 per cent less. Together the three properties raise the explained share from 42 to 53 per cent. None of the six models has a term for any of them. The many other variables offered to the random forest, among them defoliation, damage to the trees, epiphytic lichens, past management, climate, topographic wetness and nitrogen deposition, add nothing measurable once the stand and the soil are accounted for. The cover of the understorey vegetation measured in 2023 adds nothing either: converted to litter with the cover-to-biomass relations of Muukkonen et al. (2006) and the turnover rates of Liski et al. (2006), the understorey would raise the tree litter input of the plots by about 13 per cent on average, and neither the total nor the mosses and dwarf shrubs that make up most of it are related to the residual; only the cover of herbs and grasses shows a weak positive association.

The second lead of the random forest, the development class of the stand, is a measure of how far the stand has grown since it was established. The second figure makes that lead explicit and decomposes the basal-area part further, by asking how long before the sampling the stand was last regenerated (Figure 13). Over all observations the residual follows a clear course: the soil holds about 20 per cent less carbon than predicted in the first five years after a regeneration cut, about 55 per cent more at 20 to 30 years, and about as much as predicted, or somewhat less, in old stands. The deficit in the first years agrees with the loss of carbon from the organic layer that follows harvesting in many studies (James and Harrison, 2016). The time since the cut adds to the basal area of 1985 beyond doubt ($p = 4 \times 10^{-11}$; $p = 9 \times 10^{-4}$ when only dated cuts are used), but by itself the time since the cut explains little of the variance, because few of the observations fall in the window where the effect is large. Divided by the basal area of 1985, the course appears in one group only. In stands with less than 10 square metres of basal area per hectare in 1985, the soil holds about 50 per cent more carbon than predicted in the first ten years and about 150 per cent more at 10 to 30 years; in the two denser groups the residual hardly changes with time.

The two figures describe the same plots seen from two sides. The stands that were sparse in 1985 were young: their median age in 1985 was 19 years, and two thirds of them were between 10 and 30 years old, which is exactly the window in which the second figure peaks. The basal-area effect and the course after the cut are therefore one finding, and the plots measured in all three campaigns show what the finding is. In the sparse stands of 1985, the soil held about 150 per cent more carbon than predicted at the first sampling, about 70 per cent more in 2006 and about 30 per cent more in 2024. In the dense stands of 1985, the soil held about a third less carbon than predicted at the first sampling and about a quarter less in 2024. A discrepancy that is largest at the start and shrinks over the following decades, at about the pace at which the models turn their carbon over, is the signature of a starting state that is wrong for those plots, and not of an input that is missing throughout the simulation: an input missing throughout would keep the discrepancy constant or let it grow.

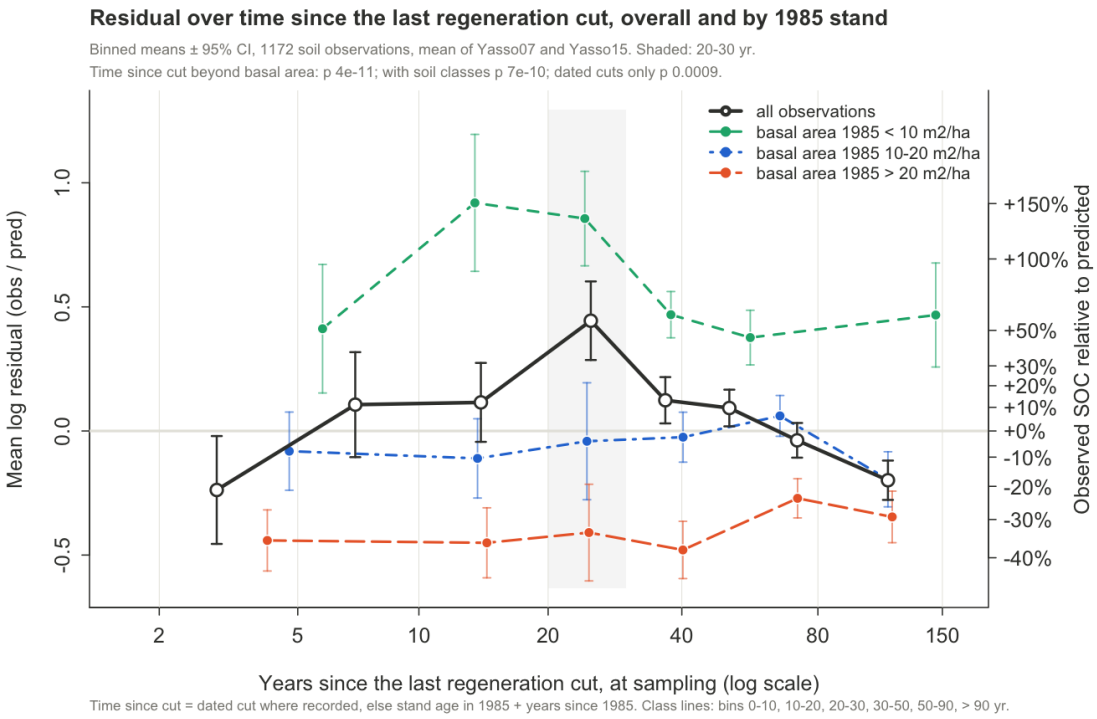


Figure 13: What the models leave behind, against the time since the last regeneration cut. The residual is defined as in Figure 12, here for each soil observation (1172 observations), with binned means and 95% confidence intervals. The time since the cut is taken from the dated cuts in the stand records and the forest use declarations where a cut precedes the sampling, and otherwise as the age of the stand in 1985 plus the years elapsed since. The solid line is all observations; the dashed lines divide them by the basal area of the stand in 1985, with wider bins. The right axis translates the residual into soil carbon relative to the prediction. The shaded band marks 20 to 30 years after the cut.

The reason the starting state is wrong for those plots lies in how the starting state is built. The path of the litter input from 1917 to 1985 has the same shape for every plot and is scaled to the litter input of each plot in 1985 (Section 2.7), so the starting state carries no information about the history of the plot. A stand regenerated in the 1960s therefore starts close to balance with the small litter input of a young stand, while the soil under it still holds much of the carbon of the stand before it and of the vegetation that followed the cut. A mature stand starts the other way round. The failure at the scale of the plot is thus the argument of this paper one level down: the starting state matters, and a starting state that does not know the history of each plot cannot reproduce the differences between plots.

Two sources can supply the carbon that the young stands of 1985 hold in excess, and the two are not exclusive. The first is the carbon left by the previous stand, which decays over decades. The second is the vegetation of early succession. After a regeneration cut, the open site is colonised first by a productive field layer and by pioneer broadleaves such as birch, rowan, aspen and willow. In Finnish Scots pine forests the photosynthesis of the ground vegetation is about eight times higher in stands 6 to 12 years old than in a stand 120 years old (Kulmala et al., 2011), and dense ground vegetation reduces the loss of carbon from a cleared site within a few years (Zehetgruber et al., 2017). Across a managed boreal landscape in northern Sweden, the carbon sink of a stand recovers within the first decade after harvest because a productive understorey establishes quickly (Peichl et al., 2022); young stands in southern Sweden turn from carbon sources into sinks 8 to 13 years after the cut (Grelle et al., 2023); and in a Finnish chronosequence the respiration of the whole ecosystem was highest in the 12-year-old stand (Kolari et al., 2004). The pioneer trees are numerous: in the young forests of Sweden the number of stems has risen since 1983, and birch accounts for most of the rise (Lidman et al., 2024). Most pioneer trees are nevertheless too small to be recorded by the inventory, which measures the diameter of a tree only once the tree is taller than 1.35 m and measures small trees on a small sub-plot only, and most are removed in cleaning ten to twenty years after the cut. Biomass models for naturally regenerated birch, aspen and rowan growing in young stands have been published only recently, those for aspen and rowan for the first time (and Repola et al., 2024), and the composition and density of seedling stands are still predicted imprecisely from inventory data (Rana et al., 2023). The litter of the pioneer trees is therefore absent from any litter product built from recorded trees, and the debris left by cleaning is not counted as harvest residue. Early-succession vegetation is a known gap in the modelling of forest carbon, and representing it with models fitted to national inventory data is the subject of ongoing work (e.g. Mehtätalo et al., 2025)[Lauri Mehtätalo (LUKE): confirm which of his publications or functions on pioneer vegetation to cite; Swedish work to extend; to discuss]. The plots cut after 1985 offer a first test, because the models simulate those cuts within the record: few of those plots are yet 20 to 30 years past the cut, and the plots that are show no comparable excess, so the data at hand cannot yet separate the two sources. A starting state that knows the history of each plot, with the cut dated from the age of the stand in 1985, is the natural next step, and the stand records needed to build it exist.

3.9 What the result may mean for soil carbon modelling and its applications

[To be written, in general terms and without naming other national inventories. Content: what follows for any application that starts a soil carbon model at steady state; reading a calibrated input multiplier as a physical quantity that can be compared with independent budgets; and the difference between skill at the scale of a national mean and skill between individual plots.]

3.10 A national map of soil carbon change as a by-product

The calibrated models also yield a gridded product. For each model and for the ensemble of the six, weighted equally, the posterior mean and standard deviation of the soil carbon stock of each plot and year were interpolated by ordinary kriging on the logarithmic scale onto a 2 km grid of the mineral-soil forest land of Finland for 1985–2024, with cells dominated by peatland or water masked. The standard deviation of each map combines the uncertainty of the interpolation with the posterior uncertainty of the model and, for the ensemble, the spread between the six models. Over 2015–2024 the ensemble gains carbon in about 87 per cent of the cells, at a mean rate of 0.18 tonnes of carbon per hectare per year. The maps are a deliverable and not a finding: the plots are on average about 27 km apart, and at that distance the stock of neighbouring plots is almost unrelated, so the maps reproduce the national field with local deviations anchored on the plots, and the detail between plots is interpolation rather than measurement. The maps are published as an open dataset [Zenodo DOI of the map deposit, once deposited]. [Before publication, check the region of carbon loss in the far south of the delta map (about −0.5 tonnes per hectare per year): a real feature of the plots, or an artefact of kriging near the edge of the grid?]

4 Limitations

The observed trend rests on three campaigns that were not carried out in the same way. The first campaign spread its fieldwork over a decade, recorded the organic layer without its litter and moss, and predicted the carbon content of most of its mineral layers from the loss of mass on ignition. We standardised what the measurements allow, but a shared offset between campaign levels cannot be separated from a change of the stock: the rate over the full observed window survives an offset of about 9 per cent, whereas the rate over the last interval, on which the apparent slowing of the accumulation rests, is removed by an offset of about 2 per cent.

The history of the soil before the first observation is represented coarsely. The path of the litter input from 1917 to 1985 is one national shape, taken from a reconstruction whose tree-derived terms cover all forest land while the area used to divide them covers upland soils only, so that the rise of the input before 1985 is, if anything, overstated. The starting state is set by one national parameter, whereas the history of exploitation that left the soil out of balance differed between regions and between plots; the models therefore cannot represent why individual plots differ, which is part of the reason for their lack of skill between plots. The starting state also has a single degree of freedom: the soil of 1917 is assumed to be in balance with the litter input of 1917, so that a soil depleted below that balance can be expressed only as a lower input, and the projected sink depends strongly on how that deficit is estimated (Section 3.6). The stock of 1917 implied by each calibration is a consequence of the calibration and is not constrained by any observation [report the implied 1917 stocks on the current run, from doublechecks/init_state_plausibility.R, and compare them with what is known of boreal forest soils; the floor for that comparison still has to be sourced].

The calibration gives every plot the same weight, although the plots in the north of the country stand for three times as much forest land as the plots in the south, because the inventory samples the north at a third of the southern density. The weights of the inventory describe how much land a plot represents, and not how much information a measurement of that plot contains. Weighting the northern plots by three would count each of them as three independent measurements, and the calibration would then claim a greater certainty about the northern soils than the 50 northern plots

can support. Treating each plot as one measurement is correct when the same parameter values hold in the north and in the south, so that the differences between the two regions are carried entirely by the climate and by the litter input. When the parameter values differ between the regions, a single calibration settles on the values suited to the south, where 88 per cent of the plots lie, and the northern plots are then fitted systematically worse.

At first sight the northern plots are indeed fitted worse: all six models predict their stocks 10 to 30 per cent too low relative to the southern plots, clearly so for Yasso15 and Yasso20. The difference, however, follows the density of the stands and not the latitude. The northern stands are sparser (10.8 against 17.0 square metres of basal area per hectare), sparse stands are underpredicted everywhere (Sections 3.7 and 3.8), and once basal area is taken into account the difference between north and south vanishes in all six models. The residuals therefore give no indication that the parameter values differ between the two regions.

The error model rests on design values. The total spread of the error, 0.800 on the logarithmic scale, is declared rather than estimated, and the spreads of the campaign offsets are prescribed rather than measured. [Decide whether to test the sensitivity of the main results to the total spread of the error (one additional set of calibrations), or to state here that it was not tested.]

The effect of estimating the starting state has not been isolated. The comparison with the published parameterisations mixes the starting state with every other difference between this calibration and the published ones, from the data to the error model, and a calibration started from a soil in balance with the present input, which would isolate the effect, was not carried out. [This paragraph limits what the title and the Conclusions may claim. Settle together with the title: run the balanced-start calibration, or phrase every causal statement about the starting state as a hypothesis.]

The projections are orders of magnitude and not forecasts. The litter input is held at its value in 2024 and the climate of the last 20 observed years is repeated, so the projections contain neither a change of climate nor a change of forest growth or management, and the projected sink is the relaxation of the soil towards balance under present conditions.

5 Conclusions

A transient starting state, estimated rather than assumed, allows all six models to represent the soil carbon measured in the three campaigns and to bracket the observed rate of change over the full observed window. The fit to these data cannot separate the six structures, and the models differ where the data do not constrain them: in the shape of their trajectories, and therefore in their extrapolation.

Under the assumptions of this study, reproducing the observed trajectory places the models at a larger litter input than the product supplies and, in two of the three versions of Yasso, at a faster turnover than the published parameterisations. The two are one displacement along a trade-off that the soil measurements cannot resolve, and moving back towards the published transit times costs little in fit, so the result suggests a direction rather than a value. The preference for a larger litter input follows the measurements more than the prior. The most plausible reading of the displacement is not that the models are wrong but that the litter input is incomplete, and the components missing from it by construction — carbon released by living roots, understorey vegetation, mycorrhizal residues and early-succession vegetation — are of the right kind, and together of a size that brackets the gap.

Where the calibration lands on the trade-off changes the projection of Yasso07 and not those of Yasso15 and Yasso20. A model that applies one climate modifier to every pool ties its turnover to its sensitivity to temperature, so that moving along the trade-off moves the projection; models that apply separate modifiers to separate pools are insensitive to the same movement, and are therefore more robust to a calibration choice that the present data cannot settle.

Under a constant litter input, the soil carbon sink of Finnish upland forests is saturating. The models place the soil 26 to 42 per cent below its state of balance, in agreement with the 38 per cent estimated by Liski et al. (2006), yet the sink they project falls by a third to a half within 60 years, because most of the remaining distance lies in the slowest pool and is realised only over centuries. How much sink remains depends on how far below balance the soil was at the start, which the calibration infers only indirectly: the soil of 1917 appears more depleted than its litter input alone would imply, and a soil assumed to start in balance with its present input would have almost nothing left to gain. A larger litter input buys additional storage, 6 to 10 tonnes of carbon per hectare by 2050 for an increase of 20 per cent, but the gain is finite, largest in the first decades, and lasts only as long as the larger input is maintained. Both statements come from projections without a change of climate and describe orders of magnitude, not forecasts.

About half of the variation between individual plots that none of the models represents can be recovered from site variables that already exist. This study is therefore a baseline for further model development: it shows where the current models fall short, in the differences between individual plots, and which information, on the productivity of the stand and on the physical properties of the soil, a next model would need to include.

Acknowledgements

[To be written. Collected so far: LUKE Data Support (weather-data provenance, 2026-09); the gridded daily weather observations are open data of the Finnish Meteorological Institute; the soil carbon measurements and their provenance (H. Ilvesniemi's workbook) go here and in the data statement, not in the references.]

Data and code availability

[To be written last. The litter product is on Zenodo; the calibration and analysis code are in the project repository and will be archived on Zenodo with a DOI at submission, so that the version used is fixed.]

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